

Thermodynamics

Instructor

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Chapter 8 – Problems

**Problem numbers correspond to
5th Edition**

8–15 The electric power needs of a community are to be met by windmills with 10-m-diameter rotors. The windmills are to be located where the wind is blowing steadily at an average velocity of 8 m/s. Determine the minimum number of windmills that need to be installed if the required power output is 600 kW.

Analysis The exergy or work potential of the blowing air is the kinetic energy it possesses,

$$\text{Exergy} = \text{ke} = \frac{V^2}{2} = \frac{(8 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) = 0.032 \text{ kJ/kg}$$

At standard atmospheric conditions (25°C, 101 kPa), the density and the mass flow rate of air are

$$\rho = \frac{P}{RT} = \frac{101 \text{ kPa}}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(298 \text{ K})} = 1.18 \text{ m}^3/\text{kg}$$

and

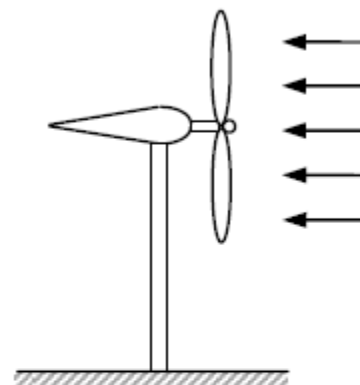
$$\dot{m} = \rho A V_1 = \rho \frac{\pi D^2}{4} V_1 = (1.18 \text{ kg/m}^3) \left(\frac{\pi}{4} \right) (10 \text{ m})^2 (8 \text{ m/s}) = 742 \text{ kg/s}$$

Thus,

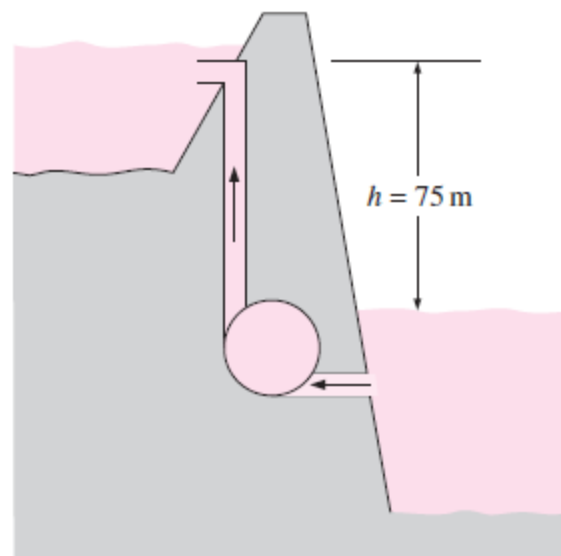
$$\text{Available Power} = \dot{m} \text{ke} = (742 \text{ kg/s})(0.032 \text{ kJ/kg}) = 23.74 \text{ kW}$$

The minimum number of windmills that needs to be installed is

$$N = \frac{\dot{W}_{\text{total}}}{\dot{W}} = \frac{600 \text{ kW}}{23.74 \text{ kW}} = 25.3 \approx \mathbf{26 \text{ windmills}}$$



8-16 One method of meeting the extra electric power demand at peak periods is to pump some water from a large body of water (such as a lake) to a water reservoir at a higher elevation at times of low demand and to generate electricity at times of high demand by letting this water run down and rotate a turbine (i.e., convert the electric energy to potential energy and then back to electric energy). For an energy storage capacity of 5×10^6 kWh, determine the minimum amount of water that needs to be stored at an average elevation (relative to the ground level) of 75 m. *Answer: 2.45×10^{10} kg*



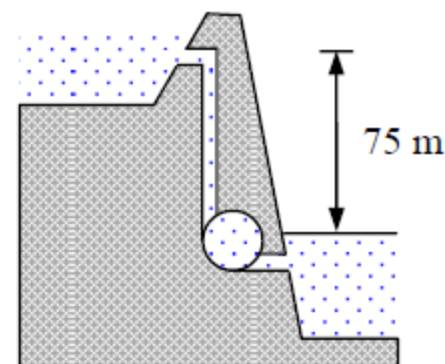
Assumptions The evaporation of water from the lake is negligible.

Analysis The exergy or work potential of the water is the potential energy it possesses,

$$\text{Exergy} = \text{PE} = mgh$$

Thus,

$$m = \frac{PE}{gh} = \frac{5 \times 10^6 \text{ kWh}}{(9.8 \text{ m/s}^2)(75 \text{ m})} \left(\frac{3600 \text{ s}}{1 \text{ h}} \right) \left(\frac{1000 \text{ m}^2/\text{s}^2}{1 \text{ kW} \cdot \text{s/kg}} \right) = 2.45 \times 10^{10} \text{ kg}$$

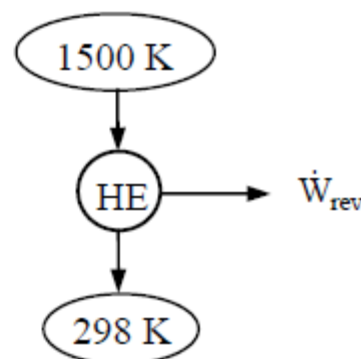


8–17 Consider a thermal energy reservoir at 1500 K that can supply heat at a rate of 150,000 kJ/h. Determine the exergy of this supplied energy, assuming an environmental temperature of 25°C.

Analysis The exergy of the supplied heat, in the rate form, is the amount of power that would be produced by a reversible heat engine,

$$\eta_{\text{th,max}} = \eta_{\text{th,rev}} = 1 - \frac{T_0}{T_H} = 1 - \frac{298 \text{ K}}{1500 \text{ K}} = 0.8013$$

$$\begin{aligned}\text{Exergy} &= \dot{W}_{\text{max,out}} = \dot{W}_{\text{rev,out}} = \eta_{\text{th,rev}} \dot{Q}_{\text{in}} \\ &= (0.8013)(150,000 / 3600 \text{ kJ/s}) \\ &= \mathbf{33.4 \text{ kW}}\end{aligned}$$



8-23 A house that is losing heat at a rate of 80,000 kJ/h when the outside temperature drops to 15°C is to be heated by electric resistance heaters. If the house is to be maintained at 22°C at all times, determine the reversible work input for this process and the irreversibility. *Answers: 0.53 kW, 21.69 kW*

Analysis The reversible work is the minimum work required to accomplish this process, and the irreversibility is the difference between the reversible work and the actual electrical work consumed. The actual power input is

$$\dot{W}_{\text{in}} = \dot{Q}_{\text{out}} = \dot{Q}_H = 80,000 \text{ kJ/h} = 22.22 \text{ kW}$$

The COP of a reversible heat pump operating between the specified temperature limits is

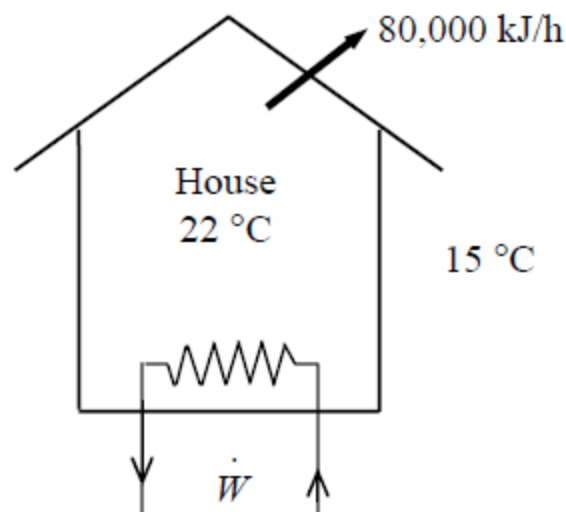
$$\text{COP}_{\text{HP,rev}} = \frac{1}{1 - T_L / T_H} = \frac{1}{1 - 288 / 295} = 42.14$$

Thus,

$$\dot{W}_{\text{rev,in}} = \frac{\dot{Q}_H}{\text{COP}_{\text{HP,rev}}} = \frac{22.22 \text{ kW}}{42.14} = 0.53 \text{ kW}$$

and

$$\dot{I} = \dot{W}_{\text{u,in}} - \dot{W}_{\text{rev,in}} = 22.22 - 0.53 = 21.69 \text{ kW}$$



8-25 Show that the power produced by a wind turbine is proportional to the cube of the wind velocity and to the square of the blade span diameter.

8-26 A geothermal power plant uses geothermal liquid water at 160°C at a rate of 440 kg/s as the heat source, and produces 14 MW of net power in an environment at 25°C. If 18.5 MW of exergy entering the plant with the geothermal water is destroyed within the plant, determine (a) the exergy of the geothermal water entering the plant, (b) the second-law efficiency, and (c) the exergy of the heat rejected from the plant.

Analysis (a) The properties of geothermal water at the inlet of the plant and at the dead state are (Tables A-4 through A-6)

$$\begin{aligned} T_1 = 160^\circ\text{C} \quad \left. \begin{array}{l} h_1 = 675.47 \text{ kJ/kg} \\ x_1 = 0 \end{array} \right\} s_1 = 1.9426 \text{ kJ/kg}\cdot\text{K} \\ T_0 = 25^\circ\text{C} \quad \left. \begin{array}{l} h_0 = 104.83 \text{ kJ/kg} \\ x_0 = 0 \end{array} \right\} s_0 = 0.36723 \text{ kJ/kg}\cdot\text{K} \end{aligned}$$

The exergy of geothermal water entering the plant is

$$\begin{aligned} \dot{X}_{\text{in}} &= \dot{m}[h_1 - h_0 - T_0(s_1 - s_0)] \\ &= (440 \text{ kg/s})[(675.47 - 104.83) \text{ kJ/kg} + 0 - (25 + 273 \text{ K})(1.9426 - 0.36723) \text{ kJ/kg}\cdot\text{K}] \\ &= 44,525 \text{ kW} = \mathbf{44.53 \text{ MW}} \end{aligned}$$

(b) The second-law efficiency of the plant is the ratio of power produced to the exergy input to the plant

$$\eta_{II} = \frac{\dot{W}_{\text{out}}}{\dot{X}_{\text{in}}} = \frac{14,000 \text{ kW}}{44,525 \text{ kW}} = \mathbf{0.314}$$

(c) The exergy of the heat rejected from the plant may be determined from an exergy balance on the plant

$$\dot{X}_{\text{heat,out}} = \dot{X}_{\text{in}} - \dot{W}_{\text{out}} - \dot{X}_{\text{dest}} = 44,525 - 14,000 - 18,500 = 12,025 \text{ kW} = \mathbf{12.03 \text{ MW}}$$

Exergy Analysis of Closed Systems

8–29 A piston–cylinder device initially contains 2 L of air at 100 kPa and 25°C. Air is now compressed to a final state of 600 kPa and 150°C. The useful work input is 1.2 kJ. Assuming the surroundings are at 100 kPa and 25°C, determine (a) the exergy of the air at the initial and the final states, (b) the minimum work that must be supplied to accomplish this compression process, and (c) the second-law efficiency of this process.

Answers: (a) 0, 0.171 kJ, (b) 0.171 kJ, (c) 14.3 percent

Analysis (a) We realize that $X_1 = \Phi_1 = 0$ since air initially is at the dead state. The mass of air is

$$m = \frac{P_1 V_1}{R T_1} = \frac{(100 \text{ kPa})(0.002 \text{ m}^3)}{(0.287 \text{ kPa} \cdot \text{m}^3 / \text{kg} \cdot \text{K})(298 \text{ K})} = 0.00234 \text{ kg}$$

Also,

$$\frac{P_2 V_2}{T_2} = \frac{P_1 V_1}{T_1} \longrightarrow V_2 = \frac{P_1 T_2}{P_2 T_1} V_1 = \frac{(100 \text{ kPa})(423 \text{ K})}{(600 \text{ kPa})(298 \text{ K})} (2 \text{ L}) = 0.473 \text{ L}$$

and

$$\begin{aligned} s_2 - s_0 &= c_{p,\text{avg}} \ln \frac{T_2}{T_0} - R \ln \frac{P_2}{P_0} \\ &= (1.009 \text{ kJ/kg} \cdot \text{K}) \ln \frac{423 \text{ K}}{298 \text{ K}} - (0.287 \text{ kJ/kg} \cdot \text{K}) \ln \frac{600 \text{ kPa}}{100 \text{ kPa}} \\ &= -0.1608 \text{ kJ/kg} \cdot \text{K} \end{aligned}$$

Thus, the exergy of air at the final state is

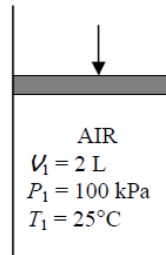
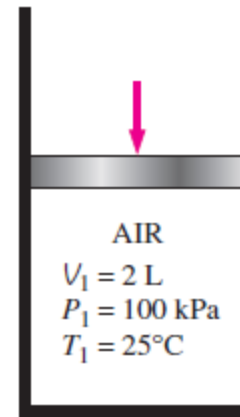
$$\begin{aligned} X_2 &= \Phi_2 = m [c_{v,\text{avg}} (T_2 - T_0) - T_0 (s_2 - s_0)] + P_0 (V_2 - V_0) \\ &= (0.00234 \text{ kg}) [(0.722 \text{ kJ/kg} \cdot \text{K})(423 - 298) \text{ K} - (298 \text{ K})(-0.1608 \text{ kJ/kg} \cdot \text{K})] \\ &\quad + (100 \text{ kPa})(0.000473 - 0.002) \text{ m}^3 [\text{kJ/m}^3 \cdot \text{kPa}] \\ &= \mathbf{0.171 \text{ kJ}} \end{aligned}$$

(b) The minimum work input is the reversible work input, which can be determined from the exergy balance by setting the exergy destruction equal to zero,

$$\begin{aligned} \underbrace{X_{\text{in}} - X_{\text{out}}}_{\text{Net exergy transfer by heat, work, and mass}} - \underbrace{X_{\text{destroyed}}}_{\text{Exergy destruction}} &= \underbrace{\Delta X_{\text{system}}}_{\text{Change in exergy}} \\ W_{\text{rev,in}} = X_2 - X_1 &= 0.171 - 0 = \mathbf{0.171 \text{ kJ}} \end{aligned}$$

(c) The second-law efficiency of this process is

$$\eta_{\text{II}} = \frac{W_{\text{rev,in}}}{W_{\text{u,in}}} = \frac{0.171 \text{ kJ}}{1.2 \text{ kJ}} = \mathbf{14.3\%}$$



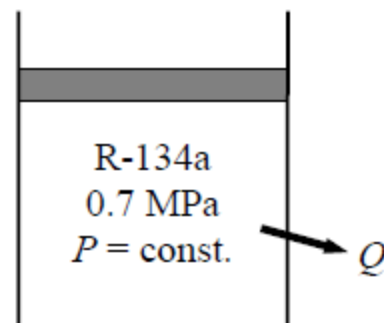
8–30 A piston–cylinder device contains 5 kg of refrigerant-134a at 0.7 MPa and 60°C. The refrigerant is now cooled at constant pressure until it exists as a liquid at 24°C. If the surroundings are at 100 kPa and 24°C, determine (a) the exergy of the refrigerant at the initial and the final states and (b) the exergy destroyed during this process.

Properties From the refrigerant tables (Tables A-11 through A-13),

$$\left. \begin{array}{l} P_1 = 0.7 \text{ MPa} \\ T_1 = 60^\circ\text{C} \end{array} \right\} \begin{array}{l} \nu_1 = 0.034875 \text{ m}^3/\text{kg} \\ u_1 = 274.01 \text{ kJ/kg} \\ s_1 = 1.0256 \text{ kJ/kg}\cdot\text{K} \end{array}$$

$$\left. \begin{array}{l} P_2 = 0.7 \text{ MPa} \\ T_2 = 24^\circ\text{C} \end{array} \right\} \begin{array}{l} \nu_2 \cong \nu_f @ 24^\circ\text{C} = 0.0008261 \text{ m}^3/\text{kg} \\ u_2 \cong u_f @ 24^\circ\text{C} = 84.44 \text{ kJ/kg} \\ s_2 \cong s_f @ 24^\circ\text{C} = 0.31958 \text{ kJ/kg}\cdot\text{K} \end{array}$$

$$\left. \begin{array}{l} P_0 = 0.1 \text{ MPa} \\ T_0 = 24^\circ\text{C} \end{array} \right\} \begin{array}{l} \nu_0 = 0.23718 \text{ m}^3/\text{kg} \\ u_0 = 251.84 \text{ kJ/kg} \\ s_0 = 1.1033 \text{ kJ/kg}\cdot\text{K} \end{array}$$



Analysis (a) From the closed system exergy relation,

$$\begin{aligned} X_1 &= \Phi_1 = m\{(u_1 - u_0) - T_0(s_1 - s_0) + P_0(\nu_1 - \nu_0)\} \\ &= (5 \text{ kg})\{(274.01 - 251.84) \text{ kJ/kg} - (297 \text{ K})(1.0256 - 1.1033) \text{ kJ/kg}\cdot\text{K} \\ &\quad + (100 \text{ kPa})(0.034875 - 0.23718) \text{ m}^3/\text{kg} \left(\frac{1 \text{ kJ}}{1 \text{ kPa}\cdot\text{m}^3} \right)\} \\ &= \mathbf{125.1 \text{ kJ}} \end{aligned}$$

and,

$$\begin{aligned}
 X_2 &= \Phi_2 = m\{(u_2 - u_0) - T_0(s_2 - s_0) + P_0(v_2 - v_0)\} \\
 &= (5 \text{ kg})\{(84.44 - 251.84) \text{ kJ/kg} - (297 \text{ K})(0.31958 - 1.1033) \text{ kJ/kg} \cdot \text{K} \\
 &\quad + (100 \text{ kPa})(0.0008261 - 0.23718) \text{ m}^3/\text{kg} \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right)\} \\
 &= \mathbf{208.6 \text{ kJ}}
 \end{aligned}$$

(b) The reversible work input, which represents the minimum work input $W_{\text{rev,in}}$ in this case can be determined from the exergy balance by setting the exergy destruction equal to zero,

$$\underbrace{X_{\text{in}} - X_{\text{out}}}_{\substack{\text{Net exergy transfer} \\ \text{by heat, work, and mass}}} - \underbrace{X_{\text{destroyed}}}_{\substack{\text{Exergy} \\ \text{destruction}}} \overset{\phi_0 \text{ (reversible)}}{=} \underbrace{\Delta X_{\text{system}}}_{\substack{\text{Change} \\ \text{in exergy}}}$$

$$W_{\text{rev,in}} = X_2 - X_1 = 208.6 - 125.1 = 83.5 \text{ kJ}$$

Noting that the process involves only boundary work, the useful work input during this process is simply the boundary work in excess of the work done by the surrounding air,

$$\begin{aligned}
 W_{\text{u,in}} &= W_{\text{in}} - W_{\text{surr,in}} = W_{\text{in}} - P_0(V_1 - V_2) = P(V_1 - V_2) - P_0m(v_1 - v_2) \\
 &= m(P - P_0)(v_1 - v_2) \\
 &= (5 \text{ kg})(700 - 100 \text{ kPa})(0.034875 - 0.0008261 \text{ m}^3/\text{kg}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) = 102.1 \text{ kJ}
 \end{aligned}$$

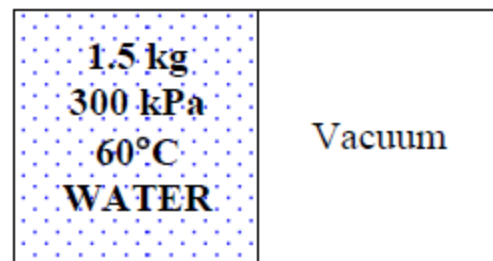
Knowing both the actual useful and reversible work inputs, the exergy destruction or irreversibility that is the difference between the two is determined from its definition to be

$$X_{\text{destroyed}} = I = W_{\text{u,in}} - W_{\text{rev,in}} = 102.1 - 83.5 = \mathbf{18.6 \text{ kJ}}$$

8-34 A rigid tank is divided into two equal parts by a partition. One part of the tank contains 1.5 kg of compressed liquid water at 300 kPa and 60°C and the other side is evacuated. Now the partition is removed, and the water expands to fill the entire tank. If the final pressure in the tank is 15 kPa, determine the exergy destroyed during this process. Assume the surroundings to be at 25°C and 100 kPa. *Answer: 3.67 kJ*

Analysis The properties of the water are (Tables A-4 through A-6)

$$\left. \begin{array}{l} P_1 = 300 \text{ kPa} \\ T_1 = 60^\circ\text{C} \end{array} \right\} \begin{array}{l} v_1 \cong v_f @ 60^\circ\text{C} = 0.001017 \text{ m}^3 / \text{kg} \\ u_1 \cong u_f @ 60^\circ\text{C} = 251.16 \text{ kJ/kg} \\ s_1 \cong s_f @ 60^\circ\text{C} = 0.8313 \text{ kJ/kg} \cdot \text{K} \end{array}$$



Noting that $v_2 = 2v_1 = 2 \times 0.001017 = 0.002034 \text{ m}^3 / \text{kg}$,

$$\left. \begin{array}{l} P_2 = 15 \text{ kPa} \\ v_2 = 0.002034 \end{array} \right\} \begin{array}{l} x_2 = \frac{v_2 - v_f}{v_{fg}} = \frac{0.002034 - 0.001014}{10.02 - 0.001014} = 0.0001017 \\ u_2 = u_f + x_2 u_{fg} = 225.93 + 0.0001017 \times 2222.1 = 226.15 \text{ kJ/kg} \\ s_2 = s_f + x_2 s_{fg} = 0.7549 + 0.0001017 \times 7.2522 = 0.7556 \text{ kJ/kg} \cdot \text{K} \end{array}$$

Taking the direction of heat transfer to be *to* the tank, the energy balance on this closed system becomes

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$Q_{\text{in}} = \Delta U = m(u_2 - u_1)$$

or,

$$Q_{\text{in}} = (1.5 \text{ kg})(226.15 - 251.16) \text{ kJ/kg} = -37.51 \text{ kJ} \rightarrow Q_{\text{out}} = 37.51 \text{ kJ}$$



The irreversibility can be determined from its definition $X_{\text{destroyed}} = T_0 S_{\text{gen}}$ where the entropy generation is determined from an entropy balance on an *extended system* that includes the tank and its immediate surroundings so that the boundary temperature of the extended system is the temperature of the surroundings at all times,

$$\underbrace{S_{\text{in}} - S_{\text{out}}}_{\text{Net entropy transfer by heat and mass}} + \underbrace{S_{\text{gen}}}_{\text{Entropy generation}} = \underbrace{\Delta S_{\text{system}}}_{\text{Change in entropy}}$$

$$-\frac{Q_{\text{out}}}{T_{\text{b,out}}} + S_{\text{gen}} = \Delta S_{\text{system}} = m(s_2 - s_1)$$

$$S_{\text{gen}} = m(s_2 - s_1) + \frac{Q_{\text{out}}}{T_{\text{surr}}}$$

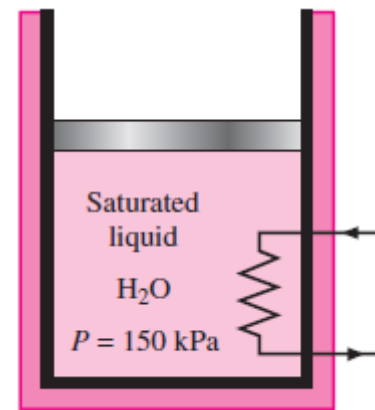
Substituting,

$$X_{\text{destroyed}} = T_0 S_{\text{gen}} = T_0 \left(m(s_2 - s_1) + \frac{Q_{\text{out}}}{T_{\text{surr}}} \right)$$

$$= (298 \text{ K}) \left[(1.5 \text{ kg})(0.7556 - 0.8313) \text{ kJ/kg} \cdot \text{K} + \frac{37.51 \text{ kJ}}{298 \text{ K}} \right]$$

$$= \mathbf{3.67 \text{ kJ}}$$

8-36 An insulated piston–cylinder device contains 2 L of saturated liquid water at a constant pressure of 150 kPa. An electric resistance heater inside the cylinder is turned on, and electrical work is done on the water in the amount of 2200 kJ. Assuming the surroundings to be at 25°C and 100 kPa, determine (a) the minimum work with which this process could be accomplished and (b) the exergy destroyed during this process. *Answers: (a) 437.7 kJ, (b) 1705 kJ*



8–38 An insulated piston–cylinder device contains 0.05 m^3 of saturated refrigerant-134a vapor at 0.8 MPa pressure. The refrigerant is now allowed to expand in a reversible manner until the pressure drops to 0.2 MPa . Determine the change in the exergy of the refrigerant during this process and the reversible work. Assume the surroundings to be at 25°C and 100 kPa .

Analysis This is a reversible adiabatic (i.e., isentropic) process, and thus $s_2 = s_1$. From the refrigerant tables (Tables A-11 through A-13),

$$P_1 = 0.8 \text{ MPa} \left\{ \begin{array}{l} v_1 = v_g @ 0.8 \text{ MPa} = 0.02562 \text{ m}^3 / \text{kg} \\ u_1 = u_g @ 0.8 \text{ MPa} = 246.79 \text{ kJ/kg} \\ s_1 = s_g @ 0.8 \text{ MPa} = 0.9183 \text{ kJ/kg} \cdot \text{K} \end{array} \right.$$

The mass of the refrigerant is

$$m = \frac{V}{v_1} = \frac{0.05 \text{ m}^3}{0.02562 \text{ m}^3 / \text{kg}} = 1.952 \text{ kg}$$

$$P_2 = 0.2 \text{ MPa} \left\{ \begin{array}{l} x_2 = \frac{s_2 - s_f}{s_{fg}} = \frac{0.9183 - 0.15457}{0.78316} = 0.9753 \\ v_2 = v_f + x_2 v_{fg} = 0.0007533 + 0.999867 \times (0.099867 - 0.0007533) = 0.09741 \text{ m}^3 / \text{kg} \\ u_2 = u_f + x_2 u_{fg} = 38.28 + 0.9753 \times 186.21 = 219.88 \text{ kJ/kg} \end{array} \right.$$

The reversible work output, which represents the maximum work output $W_{\text{rev,out}}$ can be determined from the exergy balance by setting the exergy destruction equal to zero,

$$\underbrace{X_{\text{in}} - X_{\text{out}}}_{\text{Net exergy transfer by heat, work, and mass}} - \underbrace{X_{\text{destroyed}}}_{\text{Exergy destruction}} \overset{\phi^0 \text{ (reversible)}}{=} \underbrace{\Delta X_{\text{system}}}_{\text{Change in exergy}}$$

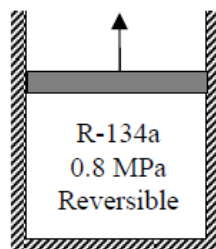
$$-W_{\text{rev,out}} = X_2 - X_1$$

$$W_{\text{rev,out}} = X_1 - X_2$$

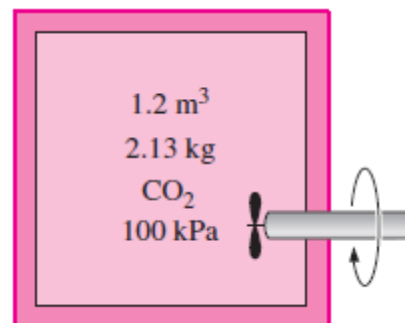
$$= \Phi_1 - \Phi_2$$

Therefore, the change in exergy and the reversible work are identical in this case. Using the definition of the closed system exergy and substituting, the reversible work is determined to be

$$\begin{aligned} W_{\text{rev,out}} &= \Phi_1 - \Phi_2 = m \left[(u_1 - u_2) - T_0 (s_1 - s_2) + P_0 (v_1 - v_2) \right] = m \left[(u_1 - u_2) + P_0 (v_1 - v_2) \right] \\ &= (1.952 \text{ kg}) [(246.79 - 219.88) \text{ kJ/kg} + (100 \text{ kPa})(0.02562 - 0.09741) \text{ m}^3 / \text{kg} [\text{kJ/kPa} \cdot \text{m}^3]] \\ &= 38.5 \text{ kJ} \end{aligned}$$



8–40 A 1.2-m^3 insulated rigid tank contains 2.13 kg of carbon dioxide at 100 kPa . Now paddle-wheel work is done on the system until the pressure in the tank rises to 120 kPa . Determine (a) the actual paddle-wheel work done during this process and (b) the minimum paddle-wheel work with which this process (between the same end states) could be accomplished. Take $T_0 = 298\text{ K}$. *Answers: (a) 87.0 kJ , (b) 7.74 kJ*

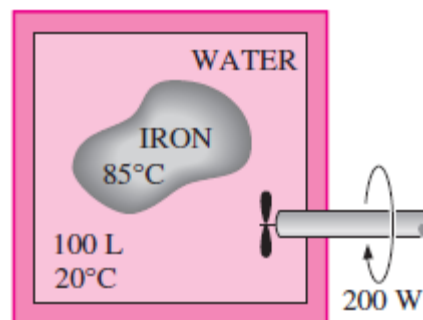


8–41 An insulated piston–cylinder device initially contains 30 L of air at 120 kPa and 27°C . Air is now heated for 5 min by a 50-W resistance heater placed inside the cylinder. The pressure of air is maintained constant during this process, and the surroundings are at 27°C and 100 kPa . Determine the exergy destroyed during this process. *Answer: 9.9 kJ*

8–42 A mass of 8 kg of helium undergoes a process from an initial state of $3\text{ m}^3/\text{kg}$ and 15°C to a final state of $0.5\text{ m}^3/\text{kg}$ and 80°C . Assuming the surroundings to be at 25°C and 100 kPa , determine the increase in the useful work potential of the helium during this process.

8–43 An insulated rigid tank is divided into two equal parts by a partition. Initially, one part contains 3 kg of argon gas at 300 kPa and 70°C , and the other side is evacuated. The partition is now removed, and the gas fills the entire tank. Assuming the surroundings to be at 25°C , determine the exergy destroyed during this process. *Answer: 129 kJ*

8-45 An iron block of unknown mass at 85°C is dropped into an insulated tank that contains 100 L of water at 20°C . At the same time, a paddle wheel driven by a 200-W motor is activated to stir the water. It is observed that thermal equilibrium is established after 20 min with a final temperature of 24°C . Assuming the surroundings to be at 20°C , determine (a) the mass of the iron block and (b) the exergy destroyed during this process. *Answers: (a) 52.0 kg, (b) 375 kJ*

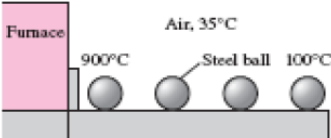


8-46 A 50-kg iron block and a 20-kg copper block, both initially at 80°C , are dropped into a large lake at 15°C . Thermal equilibrium is established after a while as a result of heat transfer between the blocks and the lake water. Assuming the surroundings to be at 20°C , determine the amount of work that could have been produced if the entire process were executed in a reversible manner.

8-48 Chickens with an average mass of 2.2 kg and average specific heat of $3.54 \text{ kJ/kg} \cdot ^{\circ}\text{C}$ are to be cooled by chilled water that enters a continuous-flow-type immersion chiller at 0.5°C and leaves at 2.5°C . Chickens are dropped into the chiller at a uniform temperature of 15°C at a rate of 500 chickens per hour and are cooled to an average temperature of 3°C before they are taken out. The chiller gains heat from the surroundings at a rate of 200 kJ/h. Determine (a) the rate of heat removal from the chicken, in kW, and (b) the rate of exergy destruction during this chilling process. Take $T_0 = 25^{\circ}\text{C}$.

8–51 Carbon steel balls ($\rho = 7833 \text{ kg/m}^3$ and $c_p = 0.465 \text{ kJ/kg} \cdot ^\circ\text{C}$) 8 mm in diameter are annealed by heating them first to 900°C in a furnace and then allowing them to cool slowly to 100°C in ambient air at 35°C . If 1200 balls are to be annealed per hour, determine (a) the rate of heat transfer from the balls to the air and (b) the rate of exergy destruction due to heat loss from the balls to the air. **Answers: (a) 260 W, (b) 146 W**

Analysis (a) We take a single ball as the system. The energy balance for this closed system can be expressed as

$$\begin{aligned} \underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} &= \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}} \\ -Q_{\text{out}} &= \Delta U_{\text{ball}} = m(u_2 - u_1) \\ Q_{\text{out}} &= mc(T_1 - T_2) \end{aligned}$$


The amount of heat transfer from a single ball is

$$\begin{aligned} m &= \rho V = \rho \frac{\pi D^3}{6} = (7833 \text{ kg/m}^3) \frac{\pi (0.008 \text{ m})^3}{6} = 0.00210 \text{ kg} \\ Q_{\text{out}} &= mc_p(T_1 - T_2) = (0.0021 \text{ kg})(0.465 \text{ kJ/kg} \cdot ^\circ\text{C})(900 - 100)^\circ\text{C} = 781 \text{ J} = 0.781 \text{ kJ (per ball)} \end{aligned}$$

Then the total rate of heat transfer from the balls to the ambient air becomes

$$\dot{Q}_{\text{out}} = \dot{n}_{\text{ball}} Q_{\text{out}} = (1200 \text{ balls/h}) \times (0.781 \text{ kJ/ball}) = 936 \text{ kJ/h} = \mathbf{260 \text{ W}}$$

(b) The exergy destruction (or irreversibility) can be determined from its definition $X_{\text{destroyed}} = T_0 S_{\text{gen}}$. The entropy generated during this process can be determined by applying an entropy balance on an *extended system* that includes the ball and its immediate surroundings so that the boundary temperature of the extended system is at 35°C at all times:

$$\begin{aligned} \underbrace{S_{\text{in}} - S_{\text{out}}}_{\text{Net entropy transfer by heat and mass}} + \underbrace{S_{\text{gen}}}_{\text{Entropy generation}} &= \underbrace{\Delta S_{\text{system}}}_{\text{Change in entropy}} \\ -\frac{Q_{\text{out}}}{T_b} + S_{\text{gen}} &= \Delta S_{\text{system}} \rightarrow S_{\text{gen}} = \frac{Q_{\text{out}}}{T_b} + \Delta S_{\text{system}} \end{aligned}$$

where

$$\Delta S_{\text{system}} = m(s_2 - s_1) = mc_{\text{avg}} \ln \frac{T_2}{T_1} = (0.00210 \text{ kg})(0.465 \text{ kJ/kg} \cdot \text{K}) \ln \frac{100 + 273}{900 + 273} = -0.00112 \text{ kJ/K}$$

Substituting,

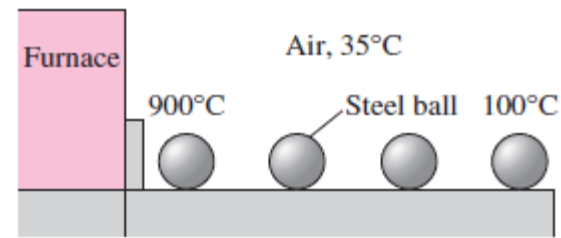
$$S_{\text{gen}} = \frac{Q_{\text{out}}}{T_b} + \Delta S_{\text{system}} = \frac{0.781 \text{ kJ}}{308 \text{ K}} - 0.00112 \text{ kJ/K} = 0.00142 \text{ kJ/K (per ball)}$$

Then the rate of entropy generation becomes

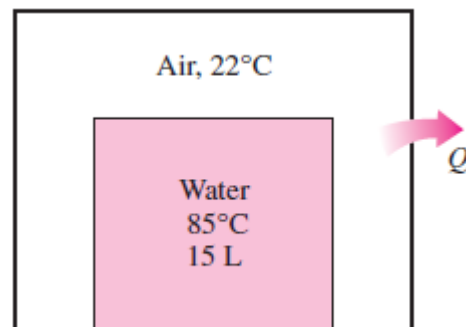
$$\dot{S}_{\text{gen}} = S_{\text{gen}} \dot{n}_{\text{ball}} = (0.00142 \text{ kJ/K} \cdot \text{ball})(1200 \text{ balls/h}) = 1.704 \text{ kJ/h} \cdot \text{K} = 0.000473 \text{ kW/K}$$

Finally,

$$\dot{X}_{\text{destroyed}} = T_0 \dot{S}_{\text{gen}} = (308 \text{ K})(0.000473 \text{ kW/K}) = 0.146 \text{ kW} = \mathbf{146 \text{ W}}$$



8-52 A 0.04-m^3 tank initially contains air at ambient conditions of 100 kPa and 22°C . Now, a 15-liter tank containing liquid water at 85°C is placed into the tank without causing any air to escape. After some heat transfer from the water to the air and the surroundings, both the air and water are measured to be at 44°C . Determine (a) the amount of heat lost to the surroundings and (b) the exergy destruction during this process.

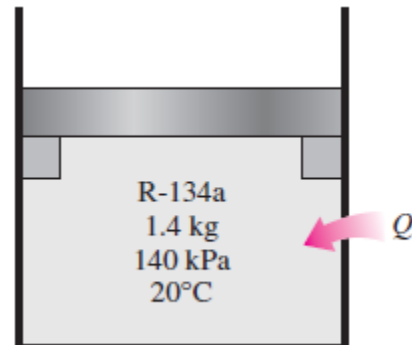


8-53 A piston–cylinder device initially contains 1.4 kg of refrigerant-134a at 140 kPa and 20°C . Heat is now transferred to the refrigerant, and the piston, which is resting on a set of stops, starts moving when the pressure inside reaches 180 kPa. Heat transfer continues until the temperature reaches 120°C . Assuming the surroundings to be at 25°C and 100 kPa, determine (a) the work done, (b) the heat transfer, (c) the exergy destroyed, and (d) the second-law efficiency of this process.

Answers: (a) 2.57 kJ, (b) 120 kJ, (c) 13.5 kJ, (d) 0.078

Exergy Analysis of Control Volumes

8-54 Steam is throttled from 8 MPa and 450°C to 6 MPa. Determine the wasted work potential during this throttling process. Assume the surroundings to be at 25°C . *Answer: 36.6 kJ/kg*





Air is compressed steadily by an 8-kW compressor from 100 kPa and 17°C to 600 kPa and 167°C at a rate of 2.1 kg/min. Neglecting the changes in kinetic and potential energies, determine (a) the increase in the exergy of the air and (b) the rate of exergy destroyed during this process. Assume the surroundings to be at 17°C.

Properties The gas constant of air is $R = 0.287 \text{ kJ/kg} \cdot \text{K}$ (Table A-1).
From the air table (Table A-17)

$$\begin{aligned} T_1 = 290 \text{ K} &\longrightarrow h_1 = 290.16 \text{ kJ/kg} \\ &\quad s_1^0 = 1.66802 \text{ kJ/kg} \cdot \text{K} \\ T_2 = 440 \text{ K} &\longrightarrow h_2 = 441.61 \text{ kJ/kg} \\ &\quad s_2^0 = 2.0887 \text{ kJ/kg} \cdot \text{K} \end{aligned}$$

Analysis The increase in exergy is the difference between the exit and inlet flow exergies,

$$\begin{aligned} \text{Increase in exergy} &= \psi_2 - \psi_1 \\ &= [(h_2 - h_1) + \Delta ke^{\text{ex}} + \Delta pe^{\text{ex}} - T_0(s_2 - s_1)] \\ &= (h_2 - h_1) - T_0(s_2 - s_1) \end{aligned}$$

where

$$\begin{aligned} s_2 - s_1 &= (s_2^0 - s_1^0) - R \ln \frac{P_2}{P_1} \\ &= (2.0887 - 1.66802) \text{ kJ/kg} \cdot \text{K} - (0.287 \text{ kJ/kg} \cdot \text{K}) \ln \frac{600 \text{ kPa}}{100 \text{ kPa}} \\ &= -0.09356 \text{ kJ/kg} \cdot \text{K} \end{aligned}$$

Substituting,

$$\begin{aligned} \text{Increase in exergy} &= \psi_2 - \psi_1 \\ &= [(441.61 - 290.16) \text{ kJ/kg} - (290 \text{ K})(-0.09356 \text{ kJ/kg} \cdot \text{K})] \\ &= \mathbf{178.6 \text{ kJ/kg}} \end{aligned}$$

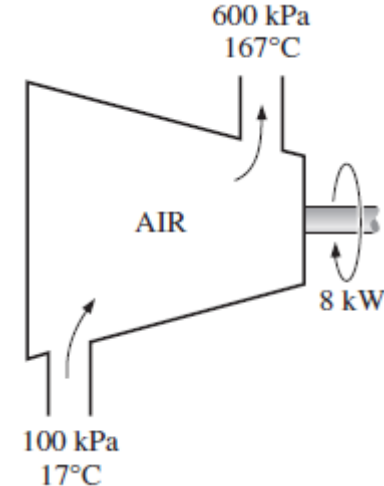
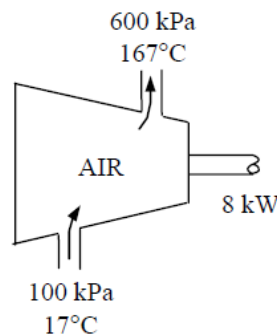
Then the reversible power input becomes

$$\dot{W}_{\text{rev, in}} = \dot{m}(\psi_2 - \psi_1) = (2.1 / 60 \text{ kg/s})(178.6 \text{ kJ/kg}) = \mathbf{6.25 \text{ kW}}$$

(b) The rate of exergy destruction (or irreversibility) is determined from its definition,

$$\dot{X}_{\text{destroyed}} = \dot{W}_{\text{in}} - \dot{W}_{\text{rev, in}} = 8 - 6.25 = \mathbf{1.75 \text{ kW}}$$

Discussion Note that 1.75 kW of power input is wasted during this compression process.



8-59 Air enters a nozzle steadily at 300 kPa and 87°C with a velocity of 50 m/s and exits at 95 kPa and 300 m/s. The heat loss from the nozzle to the surrounding medium at 17°C is estimated to be 4 kJ/kg. Determine (a) the exit temperature and (b) the exergy destroyed during this process. *Answers: (a) 39.5°C, (b) 58.4 kJ/kg*

8-61 Steam enters a diffuser at 10 kPa and 50°C with a velocity of 300 m/s and exits as saturated vapor at 50°C and 70 m/s. The exit area of the diffuser is 3 m². Determine (a) the mass flow rate of the steam and (b) the wasted work potential during this process. Assume the surroundings to be at 25°C.

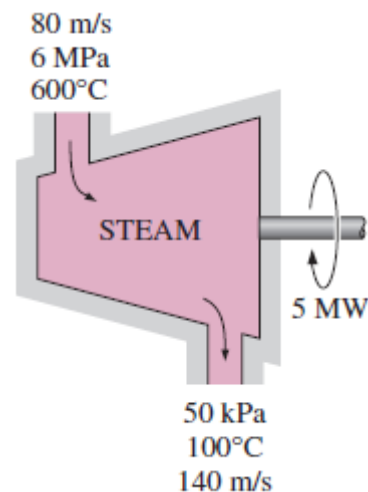
8-63 Steam enters an adiabatic turbine at 6 MPa, 600°C, and 80 m/s and leaves at 50 kPa, 100°C, and 140 m/s. If the power output of the turbine is 5 MW, determine (a) the reversible power output and (b) the second-law efficiency of the turbine. Assume the surroundings to be at 25°C.

Answers: (a) 5.84 MW, (b) 85.6 percent

8-64 Steam is throttled from 9 MPa and 500°C to a pressure of 7 MPa. Determine the decrease in exergy of the steam during this process. Assume the surroundings to be at 25°C.

Answer: 32.3 kJ/kg

8-65 Combustion gases enter a gas turbine at 900°C, 800 kPa, and 100 m/s and leave at 650°C, 400 kPa, and 220 m/s. Taking $c_p = 1.15 \text{ kJ/kg} \cdot ^\circ\text{C}$ and $k = 1.3$ for the combustion gases, determine (a) the exergy of the combustion gases at the turbine inlet and (b) the work output of the turbine under reversible conditions. Assume the surroundings to be at 25°C and 100 kPa. Can this turbine be adiabatic?



8–67 Refrigerant-134a at 140 kPa and -10°C is compressed by an adiabatic 0.5-kW compressor to an exit state of 700 kPa and 60°C . Neglecting the changes in kinetic and potential energies and assuming the surroundings to be at 27°C , determine (a) the isentropic efficiency and (b) the second-law efficiency of the compressor.

Properties The constant pressure specific heat and the specific heat ratio are given to be $c_p = 1.15 \text{ kJ/kg}\cdot\text{K}$ and $k = 1.3$. The gas constant R is determined from

$$R = c_p - c_v = c_p - c_p / k = c_p (1 - 1/k) = (1.15 \text{ kJ/kg}\cdot\text{K})(1 - 1/1.3) = 0.265 \text{ kJ/kg}\cdot\text{K}$$

Analysis (a) The exergy of the gases at the turbine inlet is simply the flow exergy,

$$\psi_1 = h_1 - h_0 - T_0(s_1 - s_0) + \frac{V_1^2}{2} + g z_1^{\phi_0}$$

where

$$\begin{aligned} s_1 - s_0 &= c_p \ln \frac{T_1}{T_0} - R \ln \frac{P_1}{P_0} \\ &= (1.15 \text{ kJ/kg}\cdot\text{K}) \ln \frac{1173 \text{ K}}{298 \text{ K}} - (0.265 \text{ kJ/kg}\cdot\text{K}) \ln \frac{800 \text{ kPa}}{100 \text{ kPa}} \\ &= 1.025 \text{ kJ/kg}\cdot\text{K} \end{aligned}$$

Thus,

$$\psi_1 = (1.15 \text{ kJ/kg}\cdot\text{K})(900 - 25)^{\circ}\text{C} - (298 \text{ K})(1.025 \text{ kJ/kg}\cdot\text{K}) + \frac{(100 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) = 705.8 \text{ kJ/kg}$$

(b) The reversible (or maximum) work output is determined from an exergy balance applied on the turbine and setting the exergy destruction term equal to zero,

$$\underbrace{\dot{X}_{\text{in}} - \dot{X}_{\text{out}}}_{\text{Rate of net exergy transfer by heat, work, and mass}} - \underbrace{\dot{X}_{\text{destroyed}}^{\phi_0}}_{\text{Rate of exergy destruction (reversible)}} = \underbrace{\Delta \dot{X}_{\text{system}}^{\phi_0}}_{\text{Rate of change of exergy (steady)}} = 0$$

$$\dot{X}_{\text{in}} = \dot{X}_{\text{out}}$$

$$\dot{m}\psi_1 = \dot{W}_{\text{rev,out}} + \dot{m}\psi_2$$

$$\dot{W}_{\text{rev,out}} = \dot{m}(\psi_1 - \psi_2) = \dot{m}[(h_1 - h_2) - T_0(s_1 - s_2) - \Delta ke - \Delta pe^{\phi_0}]$$

where

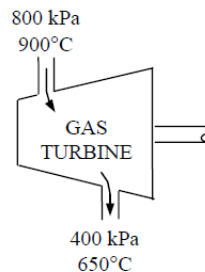
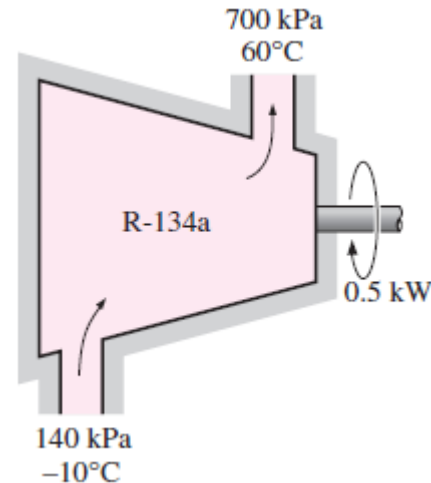
$$\Delta ke = \frac{V_2^2 - V_1^2}{2} = \frac{(220 \text{ m/s})^2 - (100 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) = 19.2 \text{ kJ/kg}$$

and

$$\begin{aligned} s_2 - s_1 &= c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \\ &= (1.15 \text{ kJ/kg}\cdot\text{K}) \ln \frac{923 \text{ K}}{1173 \text{ K}} - (0.265 \text{ kJ/kg}\cdot\text{K}) \ln \frac{400 \text{ kPa}}{800 \text{ kPa}} \\ &= -0.09196 \text{ kJ/kg}\cdot\text{K} \end{aligned}$$

Then the reversible work output on a unit mass basis becomes

$$\begin{aligned} w_{\text{rev,out}} &= h_1 - h_2 + T_0(s_2 - s_1) - \Delta ke = c_p(T_1 - T_2) + T_0(s_2 - s_1) - \Delta ke \\ &= (1.15 \text{ kJ/kg}\cdot\text{K})(900 - 650)^{\circ}\text{C} + (298 \text{ K})(-0.09196 \text{ kJ/kg}\cdot\text{K}) - 19.2 \text{ kJ/kg} = 240.9 \text{ kJ/kg} \end{aligned}$$



8–68 Air is compressed by a compressor from 95 kPa and 27°C to 600 kPa and 277°C at a rate of 0.06 kg/s. Neglecting the changes in kinetic and potential energies and assuming the surroundings to be at 25°C, determine the reversible power input for this process. *Answer: 13.7 kW*

Properties The gas constant of air is $R = 0.287 \text{ kJ/kg} \cdot \text{K}$ (Table A-1). From the air table (Table A-17)

$$T_1 = 300 \text{ K} \longrightarrow h_1 = 300.19 \text{ kJ/kg} \\ s_1^o = 1.702 \text{ kJ/kg} \cdot \text{K}$$

$$T_2 = 550 \text{ K} \longrightarrow h_2 = 555.74 \text{ kJ/kg} \\ s_2^o = 2.318 \text{ kJ/kg} \cdot \text{K}$$

Analysis The reversible (or minimum) power input is determined from the rate form of the exergy balance applied on the compressor and setting the exergy destruction term equal to zero,

$$\underbrace{\dot{X}_{\text{in}} - \dot{X}_{\text{out}}}_{\text{Rate of net exergy transfer by heat, work, and mass}} - \underbrace{\dot{X}_{\text{destroyed}}}_{\text{Rate of exergy destruction}} \overset{\text{no (reversible)}}{=} \underbrace{\Delta \dot{X}_{\text{system}}}_{\text{Rate of change of exergy}} \overset{\text{no (steady)}}{=} 0$$

$$\dot{X}_{\text{in}} = \dot{X}_{\text{out}}$$

$$\dot{m}\psi_1 + \dot{W}_{\text{rev,in}} = \dot{m}\psi_2$$

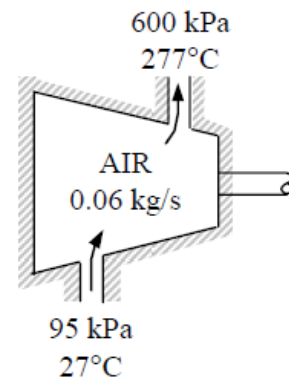
$$\dot{W}_{\text{rev,in}} = \dot{m}(\psi_2 - \psi_1) = \dot{m}[(h_2 - h_1) - T_0(s_2 - s_1) + \Delta ke^{\text{no}} + \Delta pe^{\text{no}}]$$

where

$$s_2 - s_1 = s_2^o - s_1^o - R \ln \frac{P_2}{P_1} \\ = (2.318 - 1.702) \text{ kJ/kg} \cdot \text{K} - (0.287 \text{ kJ/kg} \cdot \text{K}) \ln \frac{600 \text{ kPa}}{95 \text{ kPa}} \\ = 0.0870 \text{ kJ/kg} \cdot \text{K}$$

Substituting,

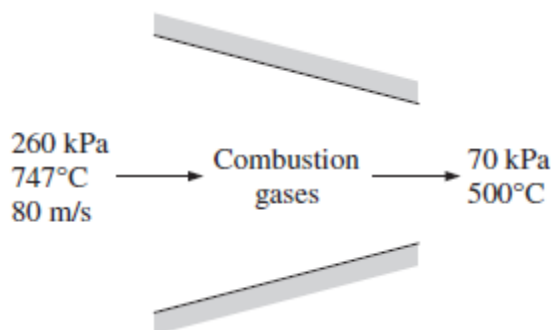
$$\dot{W}_{\text{rev,in}} = (0.06 \text{ kg/s})[(555.74 - 300.19) \text{ kJ/kg} - (298 \text{ K})(0.0870 \text{ kJ/kg} \cdot \text{K})] = \mathbf{13.7 \text{ kW}}$$



Discussion Note that a minimum of 13.7 kW of power input is needed for this compression process.

8-71 Steam expands in a turbine steadily at a rate of 15,000 kg/h, entering at 8 MPa and 450°C and leaving at 50 kPa as saturated vapor. Assuming the surroundings to be at 100 kPa and 25°C, determine (a) the power potential of the steam at the inlet conditions and (b) the power output of the turbine if there were no irreversibilities present. *Answers: (a) 5515 kW, (b) 3902 kW*

8-73 Hot combustion gases enter the nozzle of a turbojet engine at 260 kPa, 747°C, and 80 m/s and exit at 70 kPa and 500°C. Assuming the nozzle to be adiabatic and the surroundings to be at 20°C, determine (a) the exit velocity and (b) the decrease in the exergy of the gases. Take $k = 1.3$ and $c_p = 1.15 \text{ kJ/kg} \cdot ^\circ\text{C}$ for the combustion gases.



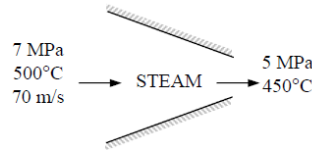
8–74 Steam is usually accelerated in the nozzle of a turbine before it strikes the turbine blades. Steam enters an adiabatic nozzle at 7 MPa and 500°C with a velocity of 70 m/s and exits at 5 MPa and 450°C. Assuming the surroundings to be at 25°C, determine (a) the exit velocity of the steam, (b) the isentropic efficiency, and (c) the exergy destroyed within the nozzle.

Properties The properties of steam at the inlet and the exit of the nozzle are (Tables A-4 through A-6)

$$\left. \begin{array}{l} P_1 = 7 \text{ MPa} \\ T_1 = 500^\circ\text{C} \end{array} \right\} \begin{array}{l} h_1 = 3411.4 \text{ kJ/kg} \\ s_1 = 6.8000 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_2 = 5 \text{ MPa} \\ T_2 = 450^\circ\text{C} \end{array} \right\} \begin{array}{l} h_2 = 3317.2 \text{ kJ/kg} \\ s_2 = 6.8210 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_{2s} = 5 \text{ MPa} \\ s_{2s} = s_1 \end{array} \right\} h_{2s} = 3302.0 \text{ kJ/kg}$$



Analysis (a) We take the nozzle to be the system, which is a control volume. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\substack{\text{Rate of net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta \dot{E}_{\text{system}}^{\text{net}}}_{\substack{\text{Rate of change in internal, kinetic,} \\ \text{potential, etc. energies}}} \stackrel{\text{net}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}(h_1 + V_1^2/2) = \dot{m}(h_2 + V_2^2/2) \quad (\text{since } \dot{W} = \dot{Q} \equiv \Delta \text{pe} \equiv 0)$$

$$0 = h_2 - h_1 + \frac{V_2^2 - V_1^2}{2}$$

Then the exit velocity becomes

$$V_2 = \sqrt{2(h_1 - h_2) + V_1^2} = \sqrt{2(3411.4 - 3317.2) \text{ kJ/kg} \left(\frac{1000 \text{ m}^2/\text{s}^2}{1 \text{ kJ/kg}} \right) + (70 \text{ m/s})^2} = 439.6 \text{ m/s}$$

(b) The exit velocity for the isentropic case is determined from

$$V_{2s} = \sqrt{2(h_1 - h_{2s}) + V_1^2} = \sqrt{2(3411.4 - 3302.0) \text{ kJ/kg} \left(\frac{1000 \text{ m}^2/\text{s}^2}{1 \text{ kJ/kg}} \right) + (70 \text{ m/s})^2} = 472.9 \text{ m/s}$$

Thus,

$$\eta_N = \frac{V_2^2/2}{V_{2s}^2/2} = \frac{(439.6 \text{ m/s})^2/2}{(472.9 \text{ m/s})^2/2} = 86.4\%$$

(c) The exergy destroyed during a process can be determined from an exergy balance or directly from its definition $X_{\text{destroyed}} = T_0 S_{\text{gen}}$ where the entropy generation S_{gen} is determined from an entropy balance on the actual nozzle. It gives

$$\underbrace{\dot{S}_{\text{in}} - \dot{S}_{\text{out}}}_{\substack{\text{Rate of net entropy transfer} \\ \text{by heat and mass}}} + \underbrace{\dot{S}_{\text{gen}}}_{\substack{\text{Rate of entropy} \\ \text{generation}}} = \underbrace{\Delta \dot{S}_{\text{system}}}_{\substack{\text{Rate of change} \\ \text{of entropy}}} \stackrel{\text{net}}{=} 0$$

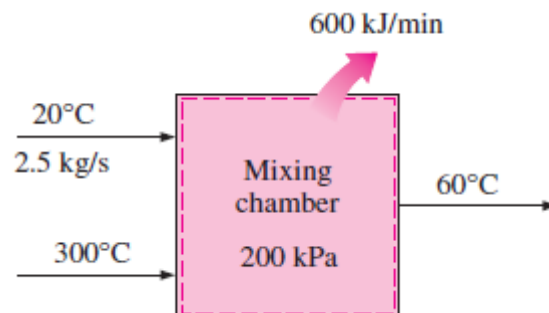
$$\dot{m}s_1 - \dot{m}s_2 + \dot{S}_{\text{gen}} = 0 \rightarrow \dot{S}_{\text{gen}} = \dot{m}(s_2 - s_1) \quad \text{or} \quad s_{\text{gen}} = s_2 - s_1$$

Substituting, the exergy destruction in the nozzle on a unit mass basis is determined to be

$$x_{\text{destroyed}} = T_0 s_{\text{gen}} = T_0 (s_2 - s_1) = (298 \text{ K})(6.8210 - 6.8000) \text{ kJ/kg} \cdot \text{K} = 6.28 \text{ kJ/kg}$$

8-75 Carbon dioxide enters a compressor at 100 kPa and 300 K at a rate of 0.2 kg/s and exits at 600 kPa and 450 K. Determine the power input to the compressor if the process involved no irreversibilities. Assume the surroundings to be at 25°C. *Answer: 25.5 kW*

8-77 Liquid water at 200 kPa and 20°C is heated in a chamber by mixing it with superheated steam at 200 kPa and 300°C. Liquid water enters the mixing chamber at a rate of 2.5 kg/s, and the chamber is estimated to lose heat to the surrounding air at 25°C at a rate of 600 kJ/min. If the mixture leaves the mixing chamber at 200 kPa and 60°C, determine (a) the mass flow rate of the superheated steam and (b) the wasted work potential during this mixing process.

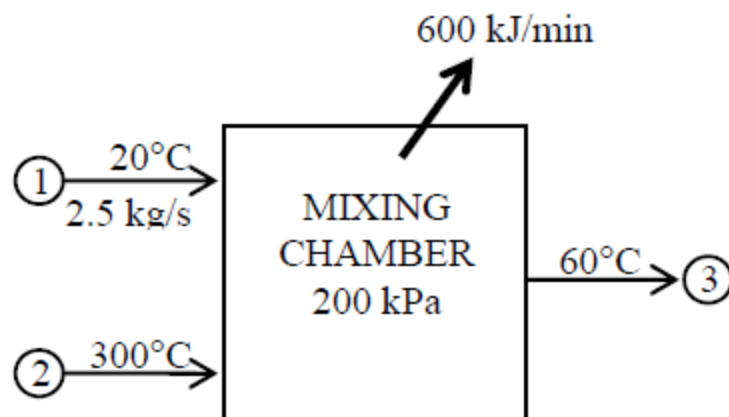


Properties Noting that $T < T_{\text{sat}} @ 200 \text{ kPa} = 120.23^\circ\text{C}$, the cold water and the exit mixture streams exist as a compressed liquid, which can be approximated as a saturated liquid at the given temperature. From Tables A-4 through A-6,

$$\left. \begin{array}{l} P_1 = 200 \text{ kPa} \\ T_1 = 20^\circ\text{C} \end{array} \right\} \begin{array}{l} h_1 \cong h_{f@20^\circ\text{C}} = 83.91 \text{ kJ/kg} \\ s_1 \cong s_{f@20^\circ\text{C}} = 0.29649 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_2 = 200 \text{ kPa} \\ T_2 = 300^\circ\text{C} \end{array} \right\} \begin{array}{l} h_2 = 3072.1 \text{ kJ/kg} \\ s_2 = 7.8941 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_3 = 200 \text{ kPa} \\ T_3 = 60^\circ\text{C} \end{array} \right\} \begin{array}{l} h_3 \cong h_{f@60^\circ\text{C}} = 251.18 \text{ kJ/kg} \\ s_3 \cong s_{f@60^\circ\text{C}} = 0.83130 \text{ kJ/kg} \cdot \text{K} \end{array}$$



Analysis (a) We take the mixing chamber as the system, which is a control volume. The mass and energy balances for this steady-flow system can be expressed in the rate form as

$$\text{Mass balance: } \dot{m}_{\text{in}} - \dot{m}_{\text{out}} = \Delta \dot{m}_{\text{system}} \stackrel{\text{no}}{\neq} 0 \text{ (steady)} = 0 \longrightarrow \dot{m}_1 + \dot{m}_2 = \dot{m}_3$$

Energy balance:

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no}}{\neq} 0 \text{ (steady)} = 0$$

$$\dot{E}_{in} = \dot{E}_{out}$$

$$\dot{m}h_1 + \dot{m}_2h_2 = \dot{Q}_{out} + \dot{m}_3h_3$$

Combining the two relations gives $\dot{Q}_{out} = \dot{m}_1h_1 + \dot{m}_2h_2 - (\dot{m}_1 + \dot{m}_2)h_3 = \dot{m}_1(h_1 - h_3) + \dot{m}_2(h_2 - h_3)$

Solving for $\dot{m}_2$ and substituting, the mass flow rate of the superheated steam is determined to be

$$\dot{m}_2 = \frac{\dot{Q}_{out} - \dot{m}_1(h_1 - h_3)}{h_2 - h_3} = \frac{(600/60 \text{ kJ/s}) - (2.5 \text{ kg/s})(83.91 - 251.18) \text{ kJ/kg}}{(3072.1 - 251.18) \text{ kJ/kg}} = \mathbf{0.148 \text{ kg/s}}$$

Also, $\dot{m}_3 = \dot{m}_1 + \dot{m}_2 = 2.5 + 0.148 = 2.648 \text{ kg/s}$

(b) The exergy destroyed during a process can be determined from an exergy balance or directly from its definition $\dot{X}_{destroyed} = T_0 \dot{S}_{gen}$ where the entropy generation $\dot{S}_{gen}$ is determined from an entropy balance on an *extended system* that includes the mixing chamber and its immediate surroundings. It gives

$$\underbrace{\dot{S}_{in} - \dot{S}_{out}}_{\text{Rate of net entropy transfer by heat and mass}} + \underbrace{\dot{S}_{gen}}_{\text{Rate of entropy generation}} = \underbrace{\Delta \dot{S}_{system}}_{\text{Rate of change of entropy}} \overset{\Phi^0}{=} 0$$

$$\dot{m}_1s_1 + \dot{m}_2s_2 - \dot{m}_3s_3 - \frac{\dot{Q}_{out}}{T_{b,surr}} + \dot{S}_{gen} = 0 \rightarrow \dot{S}_{gen} = \dot{m}_3s_3 - \dot{m}_1s_1 - \dot{m}_2s_2 + \frac{\dot{Q}_{out}}{T_0}$$

Substituting, the exergy destruction is determined to be

$$\begin{aligned} \dot{X}_{destroyed} &= T_0 \dot{S}_{gen} = T_0 \left(\dot{m}_3s_3 - \dot{m}_2s_2 - \dot{m}_1s_1 + \frac{\dot{Q}_{out}}{T_{b,surr}} \right) \\ &= (298 \text{ K})(2.648 \times 0.83130 - 0.148 \times 7.8941 - 2.5 \times 0.29649 + 10 / 298) \text{ kW/K} = \mathbf{96.4 \text{ kW}} \end{aligned}$$

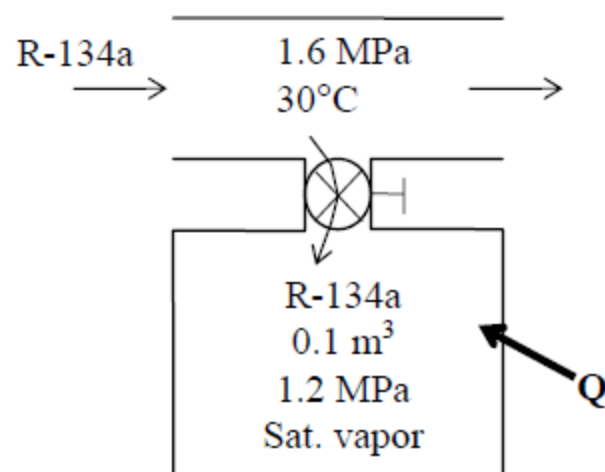
8–79 A 0.1-m³ rigid tank initially contains refrigerant-134a at 1.2 MPa and 100 percent quality. The tank is connected by a valve to a supply line that carries refrigerant-134a at 1.6 MPa and 30°C. The valve is now opened, allowing the refrigerant to enter the tank, and it is closed when the tank contains only saturated liquid at 1.4 MPa. The refrigerant exchanges heat with its surroundings at 45°C and 100 kPa during this process. Determine (a) the mass of the refrigerant that entered the tank and (b) the exergy destroyed during this process.

Properties The properties of refrigerant are (Tables A-11 through A-13)

$$\left. \begin{array}{l} P_1 = 1.2 \text{ MPa} \\ \text{sat. vapor} \end{array} \right\} \begin{array}{l} v_1 = v_g @ 1.2 \text{ MPa} = 0.01672 \text{ m}^3 / \text{kg} \\ u_1 = u_g @ 1.2 \text{ MPa} = 253.81 \text{ kJ/kg} \\ s_1 = s_g @ 1.2 \text{ MPa} = 0.91303 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_2 = 1.4 \text{ MPa} \\ \text{sat. liquid} \end{array} \right\} \begin{array}{l} v_2 = v_f @ 1.4 \text{ MPa} = 0.0009166 \text{ m}^3 / \text{kg} \\ u_2 = u_f @ 1.4 \text{ MPa} = 125.94 \text{ kJ/kg} \\ s_2 = s_f @ 1.4 \text{ MPa} = 0.45315 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_i = 1.6 \text{ MPa} \\ T_i = 30^\circ\text{C} \end{array} \right\} \begin{array}{l} h_i = 93.56 \text{ kJ/kg} \\ s_i = 0.34554 \text{ kJ/kg} \cdot \text{K} \end{array}$$



Analysis We take the tank as the system, which is a control volume. Noting that the microscopic energies of flowing and nonflowing fluids are represented by enthalpy h and internal energy u , respectively, the mass and energy balances for this uniform-flow system can be expressed as

Mass balance:

$$m_{\text{in}} - m_{\text{out}} = \Delta m_{\text{system}} \rightarrow m_i = m_2 - m_1$$

Energy balance:

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$Q_{\text{in}} + m_i h_i = m_2 u_2 - m_1 u_1 \quad (\text{since } W \cong \text{ke} \cong \text{pe} \cong 0)$$

(a) The initial and the final masses in the tank are

$$m_1 = \frac{V_1}{v_1} = \frac{0.1 \text{ m}^3}{0.01672 \text{ m}^3/\text{kg}} = 5.983 \text{ kg}$$

$$m_2 = \frac{V_2}{v_2} = \frac{0.1 \text{ m}^3}{0.0009166 \text{ m}^3/\text{kg}} = 109.10 \text{ kg}$$

Then from the mass balance

$$m_i = m_2 - m_1 = 109.10 - 5.983 = \mathbf{103.11 \text{ kg}}$$

The heat transfer during this process is determined from the energy balance to be

$$\begin{aligned} Q_{\text{in}} &= -m_i h_i + m_2 u_2 - m_1 u_1 \\ &= -(103.11 \text{ kg})(93.56 \text{ kJ/kg}) + (109.10)(125.94 \text{ kJ/kg}) - (5.983 \text{ kg})(253.81 \text{ kJ/kg}) \\ &= 2573 \text{ kJ} \end{aligned}$$



(b) The exergy destroyed during a process can be determined from an exergy balance or directly from its definition $X_{\text{destroyed}} = T_0 S_{\text{gen}}$. The entropy generation S_{gen} in this case is determined from an entropy balance on an *extended system* that includes the tank and its immediate surroundings so that the boundary temperature of the extended system is the surroundings temperature T_{surr} at all times. It gives

$$\underbrace{\frac{S_{\text{in}} - S_{\text{out}}}{T_{\text{b,in}}}}_{\text{Net entropy transfer by heat and mass}} + \underbrace{S_{\text{gen}}}_{\text{Entropy generation}} = \underbrace{\Delta S_{\text{system}}}_{\text{Change in entropy}}$$

$$\frac{Q_{\text{in}}}{T_{\text{b,in}}} + m_i s_i + S_{\text{gen}} = \Delta S_{\text{tank}} = (m_2 s_2 - m_1 s_1)_{\text{tank}} \quad \text{Substituting, the}$$

$$S_{\text{gen}} = m_2 s_2 - m_1 s_1 - m_i s_i - \frac{Q_{\text{in}}}{T_0}$$

exergy destruction is determined to be

$$\begin{aligned} X_{\text{destroyed}} &= T_0 S_{\text{gen}} = T_0 \left[m_2 s_2 - m_1 s_1 - m_i s_i - \frac{Q_{\text{in}}}{T_0} \right] \\ &= (318 \text{ K}) [109.10 \times 0.45315 - 5.983 \times 0.91303 - 103.11 \times 0.34554 - (2573 \text{ kJ}) / (318 \text{ K})] \\ &= \mathbf{80.3 \text{ kJ}} \end{aligned}$$

8–80 A 0.6-m^3 rigid tank is filled with saturated liquid water at 170°C . A valve at the bottom of the tank is now opened, and one-half of the total mass is withdrawn from the tank in liquid form. Heat is transferred to water from a source of 210°C so that the temperature in the tank remains constant. Determine (a) the amount of heat transfer and (b) the reversible work and exergy destruction for this process. Assume the surroundings to be at 25°C and 100 kPa .

Answers: (a) 2545 kJ, (b) 141.2 kJ, 141.2 kJ

8–84 A 0.2-m^3 rigid tank initially contains saturated refrigerant-134a vapor at 1 MPa . The tank is connected by a valve to a supply line that carries refrigerant-134a at 1.4 MPa and 60°C . The valve is now opened, and the refrigerant is allowed to enter the tank. The valve is closed when one-half of the volume of the tank is filled with liquid and the rest with vapor at 1.2 MPa . The refrigerant exchanges heat during this process with the surroundings at 25°C . Determine (a) the amount of heat transfer and (b) the exergy destruction associated with this process.

8–85 An insulated vertical piston–cylinder device initially contains 15 kg of water, 9 kg of which is in the vapor phase. The mass of the piston is such that it maintains a constant pressure of 200 kPa inside the cylinder. Now steam at 1 MPa and 400°C is allowed to enter the cylinder from a supply line until all the liquid in the cylinder is vaporized. Assuming the surroundings to be at 25°C and 100 kPa , determine (a) the amount of steam that has entered and (b) the exergy destroyed during this process. *Answers: (a) 23.66 kg, (b) 7610 kJ*

8–86 Consider a family of four, with each person taking a 6-minute shower every morning. The average flow rate through the shower head is 10 L/min. City water at 15°C is heated to 55°C in an electric water heater and tempered to 42°C by cold water at the T-elbow of the shower before being routed to the shower head. Determine the amount of exergy destroyed by this family per year as a result of taking daily showers. Take $T_0 = 25^\circ\text{C}$.

8–87 Ambient air at 100 kPa and 300 K is compressed isentropically in a steady-flow device to 1 MPa. Determine (a) the work input to the compressor, (b) the exergy of the air at the compressor exit, and (c) the exergy of compressed air after it is cooled to 300 K at 1 MPa pressure.

8–89 Outdoor air ($c_p = 1.005 \text{ kJ/kg} \cdot ^\circ\text{C}$) is to be preheated by hot exhaust gases in a cross-flow heat exchanger before it enters the furnace. Air enters the heat exchanger at 95 kPa and 20°C at a rate of $0.8 \text{ m}^3/\text{s}$. The combustion gases ($c_p = 1.10 \text{ kJ/kg} \cdot ^\circ\text{C}$) enter at 180°C at a rate of 1.1 kg/s and leave at 95°C . Determine the rate of heat transfer to the air and the rate of exergy destruction in the heat exchanger.

Analysis We take the exhaust pipes as the system, which is a control volume. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\approx 0 \text{ (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}h_1 = \dot{Q}_{\text{out}} + \dot{m}h_2 \quad (\text{since } \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

$$\dot{Q}_{\text{out}} = \dot{m}C_p(T_1 - T_2)$$

Then the rate of heat transfer from the exhaust gases becomes

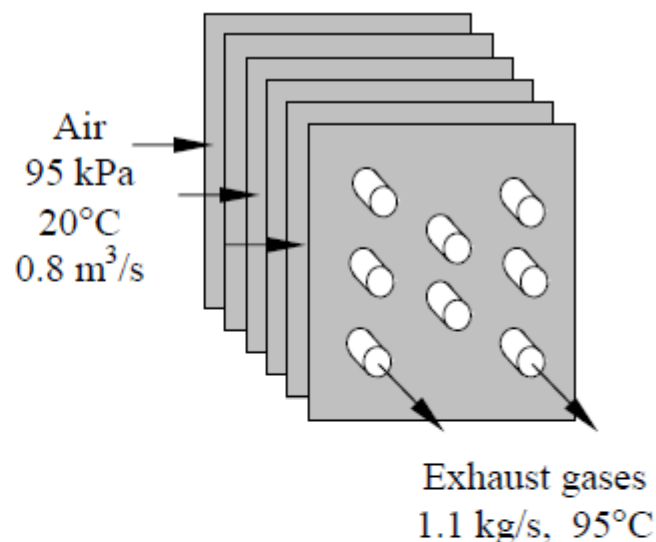
$$\dot{Q} = [\dot{m}c_p(T_{\text{in}} - T_{\text{out}})]_{\text{gas}} = (1.1 \text{ kg/s})(1.1 \text{ kJ/kg} \cdot ^\circ\text{C})(180^\circ\text{C} - 95^\circ\text{C}) = \mathbf{102.85 \text{ kW}}$$

The mass flow rate of air is

$$\dot{m} = \frac{P\dot{V}}{RT} = \frac{(95 \text{ kPa})(0.8 \text{ m}^3/\text{s})}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}) \times 293 \text{ K}} = 0.904 \text{ kg/s}$$

Noting that heat loss by exhaust gases is equal to the heat gain by the air, the air exit temperature becomes

$$\dot{Q} = [\dot{m}C_p(T_{\text{out}} - T_{\text{in}})]_{\text{air}} \rightarrow T_{\text{out}} = T_{\text{in}} + \frac{\dot{Q}}{\dot{m}c_p} = 20^\circ\text{C} + \frac{102.85 \text{ kW}}{(0.904 \text{ kg/s})(1.005 \text{ kJ/kg} \cdot ^\circ\text{C})} = 133.2^\circ\text{C}$$



The rate of entropy generation within the heat exchanger is determined by applying the rate form of the entropy balance on the entire heat exchanger:

$$\underbrace{\dot{S}_{\text{in}} - \dot{S}_{\text{out}}}_{\text{Rate of net entropy transfer by heat and mass}} + \underbrace{\dot{S}_{\text{gen}}}_{\text{Rate of entropy generation}} = \underbrace{\Delta \dot{S}_{\text{system}}}_{\text{Rate of change of entropy}} \quad \cancel{\neq 0} \text{ (steady)}$$

$$\dot{m}_1 s_1 + \dot{m}_3 s_3 - \dot{m}_2 s_2 - \dot{m}_4 s_4 + \dot{S}_{\text{gen}} = 0 \quad (\text{since } Q = 0)$$

$$\dot{m}_{\text{exhaust}} s_1 + \dot{m}_{\text{air}} s_3 - \dot{m}_{\text{exhaust}} s_2 - \dot{m}_{\text{air}} s_4 + \dot{S}_{\text{gen}} = 0$$

$$\dot{S}_{\text{gen}} = \dot{m}_{\text{exhaust}} (s_2 - s_1) + \dot{m}_{\text{air}} (s_4 - s_3)$$

Noting that both fluid streams are liquids (incompressible substances), the rate of entropy generation is

$$\begin{aligned} \dot{S}_{\text{gen}} &= \dot{m}_{\text{exhaust}} c_p \ln \frac{T_2}{T_1} + \dot{m}_{\text{air}} c_p \ln \frac{T_4}{T_3} \\ &= (1.1 \text{ kg/s})(1.1 \text{ kJ/kg.K}) \ln \frac{95 + 273}{180 + 273} + (0.904 \text{ kg/s})(1.005 \text{ kJ/kg.K}) \ln \frac{133.2 + 273}{20 + 273} = 0.0453 \text{ kW/K} \end{aligned}$$

The exergy destroyed during a process can be determined from an exergy balance or directly from its definition $X_{\text{destroyed}} = T_0 \dot{S}_{\text{gen}}$,

$$\dot{X}_{\text{destroyed}} = T_0 \dot{S}_{\text{gen}} = (293 \text{ K})(0.0453 \text{ kW/K}) = \mathbf{13.3 \text{ kW}}$$

8-90 A well-insulated shell-and-tube heat exchanger is used to heat water ($c_p = 4.18 \text{ kJ/kg} \cdot ^\circ\text{C}$) in the tubes from 20°C to 70°C at a rate of 4.5 kg/s . Heat is supplied by hot oil ($c_p = 2.30 \text{ kJ/kg} \cdot ^\circ\text{C}$) that enters the shell side at 170°C at a rate of 10 kg/s . Disregarding any heat loss from the heat exchanger, determine (a) the exit temperature of oil and (b) the rate of exergy destruction in the heat exchanger. Take $T_0 = 25^\circ\text{C}$.

Analysis We take the cold water tubes as the system, which is a control volume. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\dot{\Delta E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{Q}_{\text{in}} + \dot{m}h_1 = \dot{m}h_2 \quad (\text{since } \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

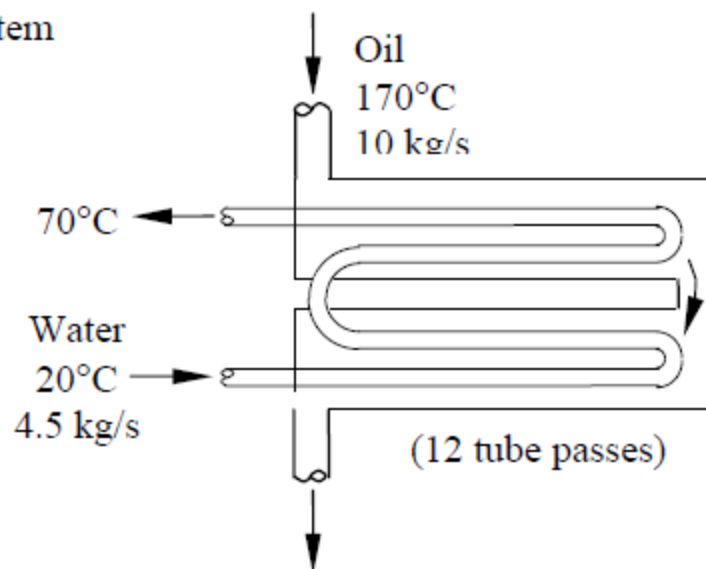
$$\dot{Q}_{\text{in}} = \dot{m}c_p(T_2 - T_1)$$

Then the rate of heat transfer to the cold water in this heat exchanger becomes

$$\dot{Q} = [\dot{m}c_p(T_{\text{out}} - T_{\text{in}})]_{\text{water}} = (4.5 \text{ kg/s})(4.18 \text{ kJ/kg} \cdot ^\circ\text{C})(70^\circ\text{C} - 20^\circ\text{C}) = 940.5 \text{ kW}$$

Noting that heat gain by the water is equal to the heat loss by the oil, the outlet temperature of the hot water is determined from

$$\dot{Q} = [\dot{m}c_p(T_{\text{in}} - T_{\text{out}})]_{\text{oil}} \rightarrow T_{\text{out}} = T_{\text{in}} - \frac{\dot{Q}}{\dot{m}c_p} = 170^\circ\text{C} - \frac{940.5 \text{ kW}}{(10 \text{ kg/s})(2.3 \text{ kJ/kg} \cdot ^\circ\text{C})} = 129.1^\circ\text{C}$$



(b) The rate of entropy generation within the heat exchanger is determined by applying the rate form of the entropy balance on the entire heat exchanger:

$$\underbrace{\dot{S}_{\text{in}} - \dot{S}_{\text{out}}}_{\text{Rate of net entropy transfer by heat and mass}} + \underbrace{\dot{S}_{\text{gen}}}_{\text{Rate of entropy generation}} = \underbrace{\Delta \dot{S}_{\text{system}}}_{\text{Rate of change of entropy}} \quad \text{0 (steady)}$$

$$\dot{m}_1 s_1 + \dot{m}_3 s_3 - \dot{m}_2 s_2 - \dot{m}_4 s_4 + \dot{S}_{\text{gen}} = 0 \quad (\text{since } Q = 0)$$

$$\dot{m}_{\text{water}} s_1 + \dot{m}_{\text{oil}} s_3 - \dot{m}_{\text{water}} s_2 - \dot{m}_{\text{oil}} s_4 + \dot{S}_{\text{gen}} = 0$$

$$\dot{S}_{\text{gen}} = \dot{m}_{\text{water}} (s_2 - s_1) + \dot{m}_{\text{oil}} (s_4 - s_3)$$

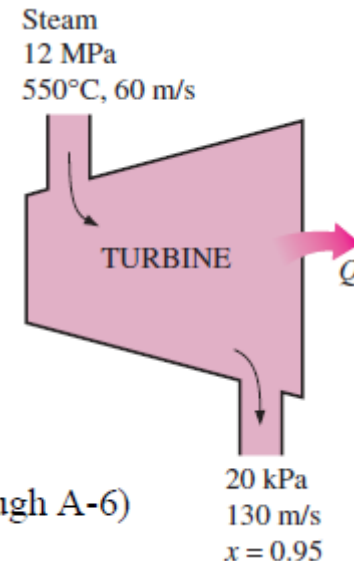
Noting that both fluid streams are liquids (incompressible substances), the rate of entropy generation is determined to be

$$\begin{aligned} \dot{S}_{\text{gen}} &= \dot{m}_{\text{water}} c_p \ln \frac{T_2}{T_1} + \dot{m}_{\text{oil}} c_p \ln \frac{T_4}{T_3} \\ &= (4.5 \text{ kg/s})(4.18 \text{ kJ/kg.K}) \ln \frac{70 + 273}{20 + 273} + (10 \text{ kg/s})(2.3 \text{ kJ/kg.K}) \ln \frac{129.1 + 273}{170 + 273} = 0.736 \text{ kW/K} \end{aligned}$$

The exergy destroyed during a process can be determined from an exergy balance or directly from its definition $X_{\text{destroyed}} = T_0 \dot{S}_{\text{gen}}$,

$$\dot{X}_{\text{destroyed}} = T_0 \dot{S}_{\text{gen}} = (298 \text{ K})(0.736 \text{ kW/K}) = \mathbf{219 \text{ kW}}$$

8-92 Steam enters a turbine at 12 MPa, 550°C, and 60 m/s and leaves at 20 kPa and 130 m/s with a moisture content of 5 percent. The turbine is not adequately insulated and it is estimated that heat is lost from the turbine at a rate of 150 kW. The power output of the turbine is 2.5 MW. Assuming the surroundings to be at 25°C, determine (a) the reversible power output of the turbine, (b) the exergy destroyed within the turbine, and (c) the second-law efficiency of the turbine. (d) Also, estimate the possible increase in the power output of the turbine if the turbine were perfectly insulated.



Analysis (a) The properties of the steam at the inlet and exit of the turbine are (Tables A-4 through A-6)

$$\left. \begin{array}{l} P_1 = 12 \text{ MPa} \\ T_1 = 550^\circ\text{C} \end{array} \right\} \begin{array}{l} h_1 = 3481.7 \text{ kJ/kg} \\ s_1 = 6.6554 \text{ kJ/kg}\cdot\text{K} \end{array}$$

$$\left. \begin{array}{l} P_2 = 20 \text{ kPa} \\ x_2 = 0.95 \end{array} \right\} \begin{array}{l} h_2 = 2491.1 \text{ kJ/kg} \\ s_2 = 7.5535 \text{ kJ/kg}\cdot\text{K} \end{array}$$

The enthalpy at the dead state is

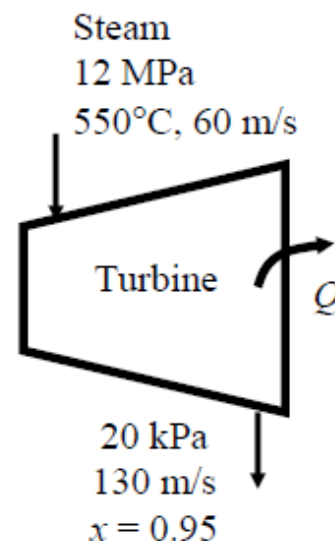
$$\left. \begin{array}{l} T_0 = 25^\circ\text{C} \\ x = 0 \end{array} \right\} h_0 = 104.83 \text{ kJ/kg}$$

The mass flow rate of steam may be determined from an energy balance on the turbine

$$\dot{m} \left(h_1 + \frac{V_1^2}{2} \right) = \dot{m} \left(h_2 + \frac{V_2^2}{2} \right) + \dot{Q}_{\text{out}} + \dot{W}_a$$

$$\dot{m} \left[3481.7 \text{ kJ/kg} + \frac{(60 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) \right] = \dot{m} \left[2491.1 \text{ kJ/kg} + \frac{(130 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) \right]$$

$$+150 \text{ kW} + 2500 \text{ kW} \longrightarrow \dot{m} = 2.693 \text{ kg/s}$$



The reversible power may be determined from

$$\begin{aligned}\dot{W}_{\text{rev}} &= \dot{m} \left[h_1 - h_2 - T_0(s_1 - s_2) + \frac{V_1^2 - V_2^2}{2} \right] \\ &= (2.693) \left[(3481.7 - 2491.1) - (298)(6.6554 - 7.5535) + \frac{(60 \text{ m/s})^2 - (130 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) \right] \\ &= \mathbf{3371 \text{ kW}}\end{aligned}$$

(b) The exergy destroyed in the turbine is

$$\dot{X}_{\text{dest}} = \dot{W}_{\text{rev}} - \dot{W}_a = 3371 - 2500 = \mathbf{871 \text{ kW}}$$

(c) The second-law efficiency is

$$\eta_{II} = \frac{\dot{W}_a}{\dot{W}_{\text{rev}}} = \frac{2500 \text{ kW}}{3371 \text{ kW}} = \mathbf{0.742}$$

(d) The energy of the steam at the turbine inlet in the given dead state is

$$\dot{Q} = \dot{m}(h_1 - h_0) = (2.693 \text{ kg/s})(3481.7 - 104.83) \text{ kJ/kg} = 9095 \text{ kW}$$

The fraction of energy at the turbine inlet that is converted to power is

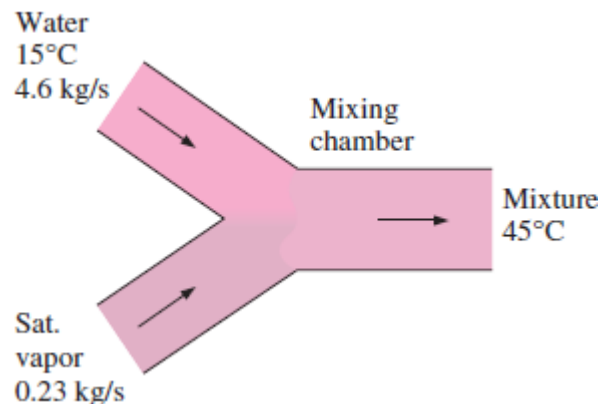
$$f = \frac{\dot{W}_a}{\dot{Q}} = \frac{2500 \text{ kW}}{9095 \text{ kW}} = 0.2749$$

Assuming that the same fraction of heat loss from the turbine could have been converted to work, the possible increase in the power if the turbine is to be well-insulated becomes

$$\dot{W}_{\text{increase}} = f\dot{Q}_{\text{out}} = (0.2749)(150 \text{ kW}) = \mathbf{41.2 \text{ kW}}$$

8-93 Air enters a compressor at ambient conditions of 100 kPa and 20°C at a rate of 4.5 m³/s with a low velocity, and exits at 900 kPa, 60°C, and 80 m/s. The compressor is cooled by cooling water that experiences a temperature rise of 10°C. The isothermal efficiency of the compressor is 70 percent. Determine (a) the actual and reversible power inputs, (b) the second-law efficiency, and (c) the mass flow rate of the cooling water.

8-94 Liquid water at 15°C is heated in a chamber by mixing it with saturated steam. Liquid water enters the chamber at the steam pressure at a rate of 4.6 kg/s and the saturated steam enters at a rate of 0.23 kg/s. The mixture leaves the mixing chamber as a liquid at 45°C. If the surroundings are at 15°C, determine (a) the temperature of saturated steam entering the chamber, (b) the exergy destruction during this mixing process, and (c) the second-law efficiency of the mixing chamber. *Answers: (a) 114.3°C, (b) 114.7 kW, (c) 0.207*

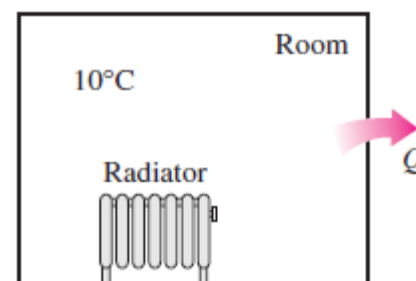


Review Problems

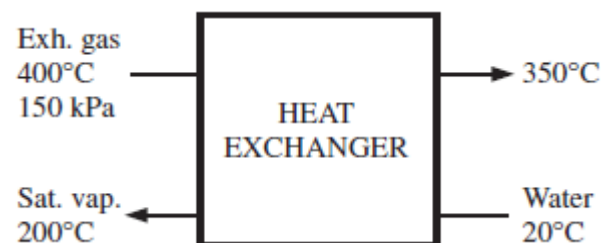
8-95 Refrigerant-134a is expanded adiabatically in an expansion valve from 1.2 MPa and 40°C to 180 kPa. For environment conditions of 100 kPa and 20°C, determine (a) the work potential of R-134a at the inlet, (b) the exergy destruction during the process, and (c) the second-law efficiency.

8-96 Steam enters an adiabatic nozzle at 3.5 MPa and 300°C with a low velocity and leaves at 1.6 MPa and 250°C at a rate of 0.4 kg/s. If the ambient state is 100 kPa and 18°C, determine (a) the exit velocity, (b) the rate of exergy destruction, and (c) the second-law efficiency.

8-97 A 30-L electrical radiator containing heating oil is placed in a well-sealed 50-m³ room. Both the air in the room and the oil in the radiator are initially at the environment temperature of 10°C. Electricity with a rating of 1.8 kW is now turned on. Heat is also lost from the room at an average rate of 0.35 kW. The heater is turned off after some time when the temperatures of the room air and oil are measured to be 20°C and 50°C, respectively. Taking the density and the specific heat of oil to be 950 kg/m³ and 2.2 kJ/kg · °C, determine (a) how long the heater is kept on, (b) the exergy destruction, and (c) the second-law efficiency for this process. *Answers: (a) 2038 s, (b) 3500 kJ, (c) 0.046*



8-98 Hot exhaust gases leaving an internal combustion engine at 400°C and 150 kPa at a rate of 0.8 kg/s is to be used to produce saturated steam at 200°C in an insulated heat exchanger. Water enters the heat exchanger at the ambient temperature of 20°C, and the exhaust gases leave the heat exchanger at 350°C. Determine (a) the rate of steam production, (b) the rate of exergy destruction in the heat exchanger, and (c) the second-law efficiency of the heat exchanger.



8-102 The inner and outer surfaces of a 0.5-cm-thick, 2-m $\times$ 2-m window glass in winter are 10°C and 3°C, respectively. If the rate of heat loss through the window is 3.2 kJ/s, determine the amount of heat loss, in kJ, through the glass over a period of 5 h. Also, determine the exergy destruction associated with this process. Take $T_0 = 5^\circ\text{C}$.

8-103 An aluminum pan has a flat bottom whose diameter is 20 cm. Heat is transferred steadily to boiling water in the pan through its bottom at a rate of 800 W. If the temperatures of the inner and outer surfaces of the bottom of the pan are 104°C and 105°C, respectively, determine the rate of exergy destruction within the bottom of the pan during this process, in W. Take $T_0 = 25^\circ\text{C}$.

8-106 Writing the first- and second-law relations and simplifying, obtain the reversible work relation for a closed system that exchanges heat with the surrounding medium at T_0 in the amount of Q_0 as well as a heat reservoir at T_R in the amount of Q_R . (*Hint:* Eliminate Q_0 between the two equations.)

8-108 Writing the first- and second-law relations and simplifying, obtain the reversible work relation for a uniform-flow system that exchanges heat with the surrounding medium at T_0 in the amount of Q_0 as well as a heat reservoir at T_R in the amount of Q_R . (Hint: Eliminate Q_0 between the two equations.)

Analysis We take the direction of heat transfers to be to the system (heat input) and the direction of work transfer to be from the system (work output). The result obtained is still general since quantities with opposite directions can be handled the same way by using negative signs. The energy and entropy balances for this stationary closed system can be expressed as

Energy balance: $E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$

$$Q_0 + Q_R - W = \sum m_e \left(h_e + \frac{V_e^2}{2} + gz_e \right) - \sum m_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) + (U_2 - U_1)_{\text{cv}}$$

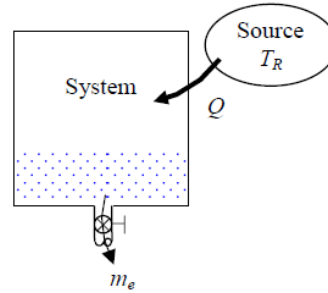
or,
$$W = \sum m_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \sum m_e \left(h_e + \frac{V_e^2}{2} + gz_e \right) - (U_2 - U_1)_{\text{cv}} + Q_0 + Q_R \quad (1)$$

Entropy balance: $S_{\text{in}} - S_{\text{out}} + S_{\text{gen}} = \Delta S_{\text{system}}$

$$S_{\text{gen}} = (S_2 - S_1)_{\text{cv}} + \sum m_e s_e - \sum m_i s_i + \frac{-Q_R}{T_R} + \frac{-Q_0}{T_0} \quad (2)$$

Solving for Q_0 from (2) and substituting in (1) yields

$$W = \sum m_i \left(h_i + \frac{V_i^2}{2} + gz_i - T_0 s_i \right) - \sum m_e \left(h_e + \frac{V_e^2}{2} + gz_e - T_0 s_e \right) + [(U_1 - U_2) - T_0 (S_1 - S_2)]_{\text{cv}} - T_0 S_{\text{gen}} - Q_R \left(1 - \frac{T_0}{T_R} \right)$$



The useful work relation for a closed system is obtained from

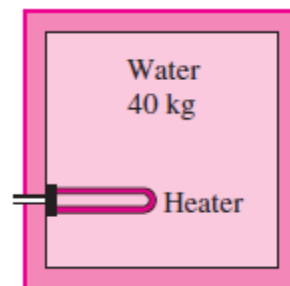
$$W_u = W - W_{\text{surr}} = \sum m_i \left(h_i + \frac{V_i^2}{2} + gz_i - T_0 s_i \right) - \sum m_e \left(h_e + \frac{V_e^2}{2} + gz_e - T_0 s_e \right) + [(U_1 - U_2) - T_0 (S_1 - S_2)]_{\text{cv}} - T_0 S_{\text{gen}} - Q_R \left(1 - \frac{T_0}{T_R} \right) - P_0 (V_2 - V_1)$$

Then the reversible work relation is obtained by substituting $S_{\text{gen}} = 0$,

$$W_{\text{rev}} = \sum m_i \left(h_i + \frac{V_i^2}{2} + gz_i - T_0 s_i \right) - \sum m_e \left(h_e + \frac{V_e^2}{2} + gz_e - T_0 s_e \right) + [(U_1 - U_2) - T_0 (S_1 - S_2) + P_0 (V_1 - V_2)]_{\text{cv}} - Q_R \left(1 - \frac{T_0}{T_R} \right)$$

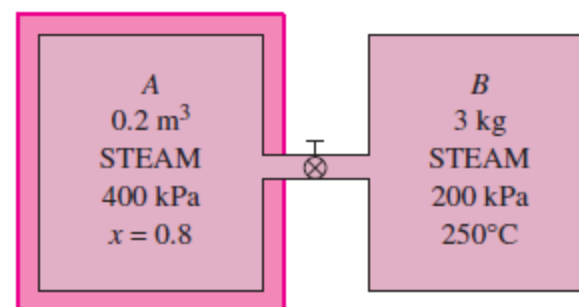
A positive result for W_{rev} indicates work output, and a negative result work input. Also, the Q_R is a positive quantity for heat transfer to the system, and a negative quantity for heat transfer from the system.

8-109 A 50-cm-long, 800-W electric resistance heating element whose diameter is 0.5 cm is immersed in 40 kg of water initially at 20°C. Assuming the water container is well-insulated, determine how long it will take for this heater to raise the water temperature to 80°C. Also, determine the minimum work input required and exergy destruction for this process, in kJ. Take $T_0 = 20^\circ\text{C}$.



8-110 A 5-cm-external-diameter, 10-m-long hot water pipe at 80°C is losing heat to the surrounding air at 5°C by natural convection at a rate of 45 W. Determine the rate at which the work potential is wasted during this process as a result of this heat loss.

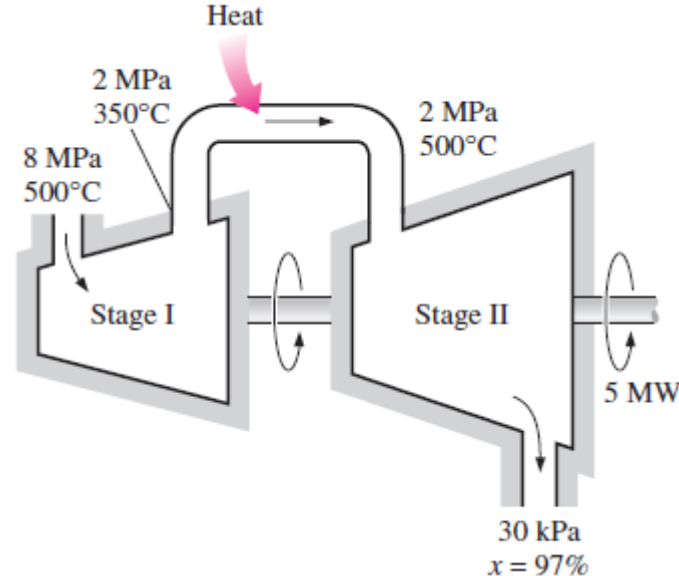
8-111 Two rigid tanks are connected by a valve. Tank *A* is insulated and contains 0.2 m³ of steam at 400 kPa and 80 percent quality. Tank *B* is uninsulated and contains 3 kg of steam at 200 kPa and 250°C. The valve is now opened, and steam flows from tank *A* to tank *B* until the pressure in tank *A* drops to 300 kPa. During this process 900 kJ of heat is transferred from tank *B* to the surroundings at 0°C. Assuming the steam remaining inside tank *A* to have undergone a reversible adiabatic process, determine (a) the final temperature in each tank and (b) the work potential wasted during this process.



8-115 Steam at 9 MPa and 500°C enters a two-stage adiabatic turbine at a rate of 15 kg/s. Ten percent of the steam is extracted at the end of the first stage at a pressure of 1.4 MPa for other use. The remainder of the steam is further expanded in the second stage and leaves the turbine at 50 kPa. If the turbine has an isentropic efficiency of 88 percent, determine the wasted power potential during this process as a result of irreversibilities. Assume the surroundings to be at 25°C.

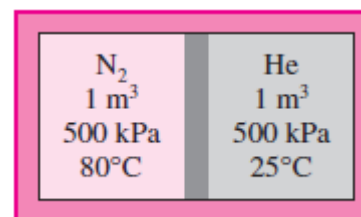
8-116 Steam enters a two-stage adiabatic turbine at 8 MPa and 500°C. It expands in the first stage to a state of 2 MPa and 350°C. Steam is then reheated at constant pressure to a temperature of 500°C before it is routed to the second stage, where it exits at 30 kPa and a quality of 97 percent. The work output of the turbine is 5 MW. Assuming the surroundings to be at 25°C, determine the reversible power output and the rate of exergy destruction within this turbine.

Answers: 5457 kW, 457 kW



8-117 One ton of liquid water at 80°C is brought into a well-insulated and well-sealed 4-m × 5-m × 6-m room initially at 22°C and 100 kPa. Assuming constant specific heats for both the air and water at room temperature, determine (a) the final equilibrium temperature in the room, (b) the exergy destruction, (c) the maximum amount of work that can be produced during this process, in kJ. Take $T_0 = 10^\circ\text{C}$.

8-118 Consider a well-insulated horizontal rigid cylinder that is divided into two compartments by a piston that is free to move but does not allow either gas to leak into the other side. Initially, one side of the piston contains 1 m³ of N₂ gas at 500 kPa and 80°C while the other side contains 1 m³ of He gas at 500 kPa and 25°C. Now thermal equilibrium is established in the cylinder as a result of heat transfer through the piston. Using constant specific heats at room temperature, determine (a) the final equilibrium temperature in the cylinder and (b) the wasted work potential during this process. What would your answer be if the piston were not free to move? Take $T_0 = 25^\circ\text{C}$.



8–119 Repeat Prob. 8–118 by assuming the piston is made of 5 kg of copper initially at the average temperature of the two gases on both sides.

Analysis The mass of each gas in the cylinder is

$$m_{\text{N}_2} = \left(\frac{P_1 V_1}{RT_1} \right)_{\text{N}_2} = \frac{(500 \text{ kPa})(1 \text{ m}^3)}{(0.2968 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(353 \text{ K})} = 4.77 \text{ kg}$$

$$m_{\text{He}} = \left(\frac{P_1 V_1}{RT_1} \right)_{\text{He}} = \frac{(500 \text{ kPa})(1 \text{ m}^3)}{(2.0769 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(298 \text{ K})} = 0.808 \text{ kg}$$

Taking the entire contents of the cylinder as our system, the 1st law relation can be written as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$0 = \Delta U = (\Delta U)_{\text{N}_2} + (\Delta U)_{\text{He}}$$

$$0 = [mc_v(T_2 - T_1)]_{\text{N}_2} + [mc_v(T_2 - T_1)]_{\text{He}}$$

Substituting,

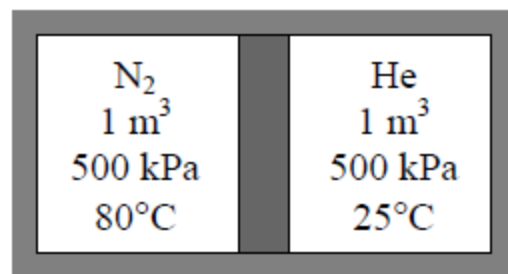
$$(4.77 \text{ kg})(0.743 \text{ kJ/kg} \cdot ^\circ\text{C})(T_f - 80)^\circ\text{C} + (0.808 \text{ kg})(3.1156 \text{ kJ/kg} \cdot ^\circ\text{C})(T_f - 25)^\circ\text{C} = 0$$

It gives

$$T_f = 57.2^\circ\text{C}$$

where T_f is the final equilibrium temperature in the cylinder.

The answer would be the **same** if the piston were not free to move since it would effect only pressure, and not the specific heats.



(b) We take the entire cylinder as our system, which is a closed system. Noting that the cylinder is well-insulated and thus there is no heat transfer, the entropy balance for this closed system can be expressed as

$$\underbrace{S_{\text{in}} - S_{\text{out}}}_{\text{Net entropy transfer by heat and mass}} + \underbrace{S_{\text{gen}}}_{\text{Entropy generation}} = \underbrace{\Delta S_{\text{system}}}_{\text{Change in entropy}}$$

$$0 + S_{\text{gen}} = \Delta S_{\text{N}_2} + \Delta S_{\text{He}}$$

But first we determine the final pressure in the cylinder:

$$N_{\text{total}} = N_{\text{N}_2} + N_{\text{He}} = \left(\frac{m}{M} \right)_{\text{N}_2} + \left(\frac{m}{M} \right)_{\text{He}} = \frac{4.77 \text{ kg}}{28 \text{ kg/kmol}} + \frac{0.808 \text{ kg}}{4 \text{ kg/kmol}} = 0.372 \text{ kmol}$$

$$P_2 = \frac{N_{\text{total}} R_u T}{V_{\text{total}}} = \frac{(0.372 \text{ kmol})(8.314 \text{ kPa} \cdot \text{m}^3/\text{kmol} \cdot \text{K})(330.2 \text{ K})}{2 \text{ m}^3} = 510.6 \text{ kPa}$$

Then,

$$\Delta S_{\text{N}_2} = m \left(c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \right)_{\text{N}_2}$$

$$= (4.77 \text{ kg}) \left[(1.039 \text{ kJ/kg} \cdot \text{K}) \ln \frac{330.2 \text{ K}}{353 \text{ K}} - (0.2968 \text{ kJ/kg} \cdot \text{K}) \ln \frac{510.6 \text{ kPa}}{500 \text{ kPa}} \right] = -0.361 \text{ kJ/K}$$

$$\Delta S_{\text{He}} = m \left(c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \right)_{\text{He}}$$

$$= (0.808 \text{ kg}) \left[(5.1926 \text{ kJ/kg} \cdot \text{K}) \ln \frac{330.2 \text{ K}}{298 \text{ K}} - (2.0769 \text{ kJ/kg} \cdot \text{K}) \ln \frac{510.6 \text{ kPa}}{500 \text{ kPa}} \right] = 0.395 \text{ kJ/K}$$

$$S_{\text{gen}} = \Delta S_{\text{N}_2} + \Delta S_{\text{He}} = -0.361 + 0.395 = 0.034 \text{ kJ/K}$$

The wasted work potential is equivalent to the exergy destroyed during a process, and it can be determined from an exergy balance or directly from its definition $X_{\text{destroyed}} = T_0 S_{\text{gen}}$,

$$X_{\text{destroyed}} = T_0 S_{\text{gen}} = (298 \text{ K})(0.034 \text{ kJ/K}) = \mathbf{10.1 \text{ kJ}}$$

If the piston were not free to move, we would still have $T_2 = 330.2 \text{ K}$ but the volume of each gas would remain constant in this case:

$$\Delta S_{\text{N}_2} = m \left(c_v \ln \frac{T_2}{T_1} - R \ln \frac{v_2}{v_1} \right)_{\text{N}_2} = (4.77 \text{ kg})(0.743 \text{ kJ/kg} \cdot \text{K}) \ln \frac{330.2 \text{ K}}{353 \text{ K}} = -0.237 \text{ kJ/K}$$

$$\Delta S_{\text{He}} = m \left(c_v \ln \frac{T_2}{T_1} - R \ln \frac{v_2}{v_1} \right)_{\text{He}} = (0.808 \text{ kg})(3.1156 \text{ kJ/kg} \cdot \text{K}) \ln \frac{330.2 \text{ K}}{298 \text{ K}} = 0.258 \text{ kJ/K}$$

$$S_{\text{gen}} = \Delta S_{\text{N}_2} + \Delta S_{\text{He}} = -0.237 + 0.258 = 0.021 \text{ kJ/K}$$

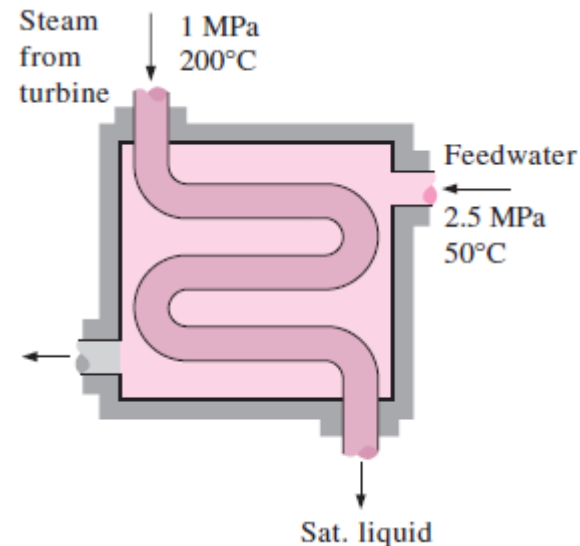
and

$$X_{\text{destroyed}} = T_0 S_{\text{gen}} = (298 \text{ K})(0.021 \text{ kJ/K}) = \mathbf{6.26 \text{ kJ}}$$

8-121

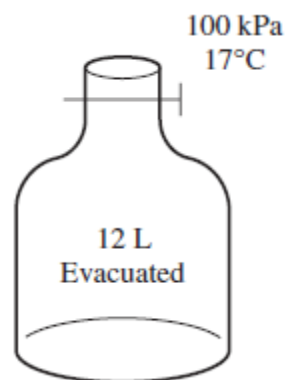
In large steam power plants, the feedwater is frequently heated in closed feedwater heaters, which are basically heat exchangers, by steam extracted from the turbine at some stage. Steam enters the feedwater heater at 1 MPa and 200°C and leaves as saturated liquid at the same pressure. Feedwater enters the heater at 2.5 MPa and 50°C and leaves 10°C below the exit temperature of the steam. Neglecting any heat losses from the outer surfaces of the heater, determine (a) the ratio of the mass flow rates of the extracted steam and the feedwater heater and (b) the reversible work for this process per unit mass of the feedwater. Assume the surroundings to be at 25°C.

Answers: (a) 0.247, (b) 63.5 kJ/kg

**8-123**

In order to cool 1 ton of water at 20°C in an insulated tank, a person pours 80 kg of ice at -5°C into the water. Determine (a) the final equilibrium temperature in the tank and (b) the exergy destroyed during this process. The melting temperature and the heat of fusion of ice at atmospheric pressure are 0°C and 333.7 kJ/kg, respectively. Take $T_0 = 20^{\circ}\text{C}$.

8-124 Consider a 12-L evacuated rigid bottle that is surrounded by the atmosphere at 100 kPa and 17°C. A valve at the neck of the bottle is now opened and the atmospheric air is allowed to flow into the bottle. The air trapped in the bottle eventually reaches thermal equilibrium with the atmosphere as a result of heat transfer through the wall of the bottle. The valve remains open during the process so that the trapped air also reaches mechanical equilibrium with the atmosphere. Determine the net heat transfer through the wall of the bottle and the exergy destroyed during this filling process.



Analysis We take the bottle as the system, which is a control volume since mass crosses the boundary. Noting that the microscopic energies of flowing and nonflowing fluids are represented by enthalpy h and internal energy u , respectively, the mass and energy balances can be expressed as

Mass balance: $m_{\text{in}} - m_{\text{out}} = \Delta m_{\text{system}} \rightarrow m_i = m_2$ (since $m_{\text{out}} = m_{\text{initial}} = 0$)

Energy balance:

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$Q_{\text{in}} + m_i h_i = m_2 u_2 \quad (\text{since } W \cong E_{\text{out}} = E_{\text{initial}} = \text{ke} \cong \text{pe} \cong 0)$$

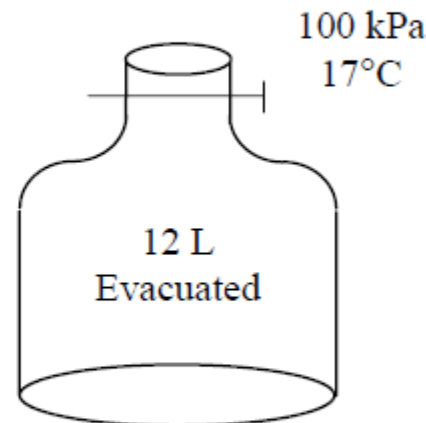
Combining the two balances:

$$Q_{\text{in}} = m_2 (u_2 - h_i)$$

where

$$m_2 = \frac{P_2 V}{RT_2} = \frac{(100 \text{ kPa})(0.012 \text{ m}^3)}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(290 \text{ K})} = 0.0144 \text{ kg}$$

$$T_i = T_2 = 290 \text{ K} \xrightarrow{\text{Table A-17}} \begin{aligned} h_i &= 290.16 \text{ kJ/kg} \\ u_2 &= 206.91 \text{ kJ/kg} \end{aligned}$$



Substituting,

$$Q_{in} = (0.0144 \text{ kg})(206.91 - 290.16) \text{ kJ/kg} = - 1.2 \text{ kJ} \rightarrow Q_{out} = \mathbf{1.2 \text{ kJ}}$$

Note that the negative sign for heat transfer indicates that the assumed direction is wrong. Therefore, we reversed the direction.

The entropy generated during this process is determined by applying the entropy balance on an *extended system* that includes the bottle and its immediate surroundings so that the boundary temperature of the extended system is the temperature of the surroundings at all times. The entropy balance for it can be expressed as

$$\underbrace{S_{in} - S_{out}}_{\substack{\text{Net entropy transfer} \\ \text{by heat and mass}}} + \underbrace{S_{gen}}_{\substack{\text{Entropy} \\ \text{generation}}} = \underbrace{\Delta S_{system}}_{\substack{\text{Change} \\ \text{in entropy}}}$$
$$m_i s_i - \frac{Q_{out}}{T_{b,in}} + S_{gen} = \Delta S_{tank} = m_2 s_2 - m_1 s_1 \stackrel{\neq 0}{=} m_2 s_2$$

Therefore, the total entropy generated during this process is

$$S_{gen} = -m_i s_i + m_2 s_2 + \frac{Q_{out}}{T_{b,out}} = m_2 (s_2 - s_i) \stackrel{\neq 0}{=} + \frac{Q_{out}}{T_{b,out}} = \frac{Q_{out}}{T_{surr}} = \frac{1.2 \text{ kJ}}{290 \text{ K}} = 0.00415 \text{ kJ/K}$$

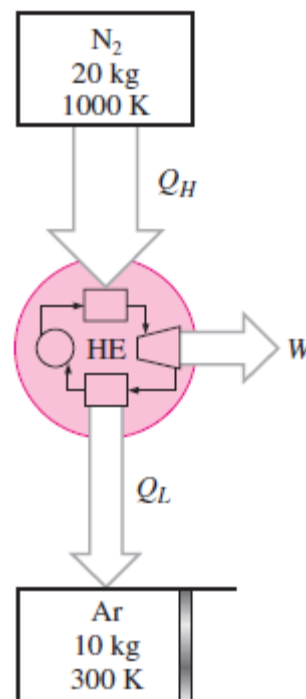
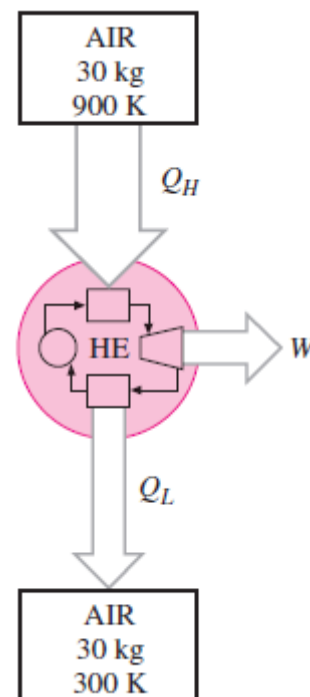
The exergy destroyed during a process can be determined from an exergy balance or directly from its definition $X_{destroyed} = T_0 S_{gen}$,

$$X_{destroyed} = T_0 S_{gen} = (290 \text{ K})(0.00415 \text{ kJ/K}) = \mathbf{1.2 \text{ kJ}}$$

8-125 Two constant-volume tanks, each filled with 30 kg of air, have temperatures of 900 K and 300 K. A heat engine placed between the two tanks extracts heat from the high-temperature tank, produces work, and rejects heat to the low-temperature tank. Determine the maximum work that can be produced by the heat engine and the final temperatures of the tanks. Assume constant specific heats at room temperature.

8-126 Two constant-pressure devices, each filled with 30 kg of air, have temperatures of 900 K and 300 K. A heat engine placed between the two devices extracts heat from the high-temperature device, produces work, and rejects heat to the low-temperature device. Determine the maximum work that can be produced by the heat engine and the final temperatures of the devices. Assume constant specific heats at room temperature.

8-129 A constant-volume tank contains 20 kg of nitrogen at 1000 K, and a constant-pressure device contains 10 kg of argon at 300 K. A heat engine placed between the tank and device extracts heat from the high-temperature tank, produces work, and rejects heat to the low-temperature device. Determine the maximum work that can be produced by the heat engine and the final temperatures of the nitrogen and argon. Assume constant specific heats at room temperature.



8-130 A constant-volume tank has a temperature of 800 K and a constant-pressure device has a temperature of 290 K. Both the tank and device are filled with 20 kg of air. A heat engine placed between the tank and device receives heat from the high-temperature tank, produces work, and rejects heat to the low-temperature device. Determine the maximum work that can be produced by the heat engine and the final temperatures of the tank and device. Assume constant specific heats at room temperature.

8-131 Can closed-system exergy be negative? How about flow exergy? Explain using an incompressible substance as an example.

Analysis The availability of a closed system cannot be negative. However, the flow availability can be negative at low pressures. A closed system has zero availability at dead state, and positive availability at any other state since we can always produce work when there is a pressure or temperature differential.

To see that the flow availability can be negative, consider an incompressible substance. The flow availability can be written as

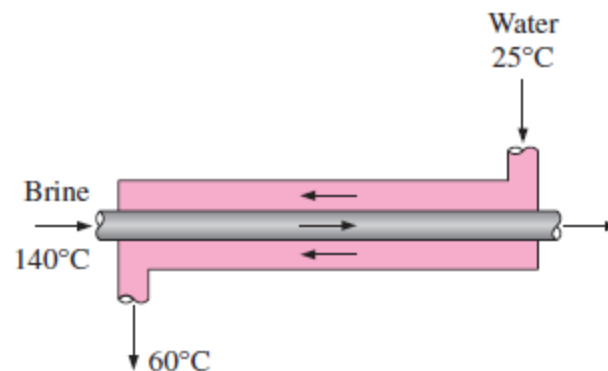
$$\begin{aligned}\psi &= h - h_0 + T_0(s - s_0) \\ &= (u - u_0) + v(P - P_0) + T_0(s - s_0) \\ &= \xi + v(P - P_0)\end{aligned}$$

The closed system availability ξ is always positive or zero, and the flow availability can be negative when $P \ll P_0$.

8-134 Long cylindrical steel rods ($\rho = 7833 \text{ kg/m}^3$ and $c_p = 0.465 \text{ kJ/kg} \cdot ^\circ\text{C}$) of 10-cm diameter are heat-treated by drawing them at a velocity of 3 m/min through a 6-m-long oven maintained at 900°C . If the rods enter the oven at 30°C and leave at 700°C , determine (a) the rate of heat transfer to the rods in the oven and (b) the rate of exergy destruction associated with this heat transfer process. Take $T_0 = 25^\circ\text{C}$.

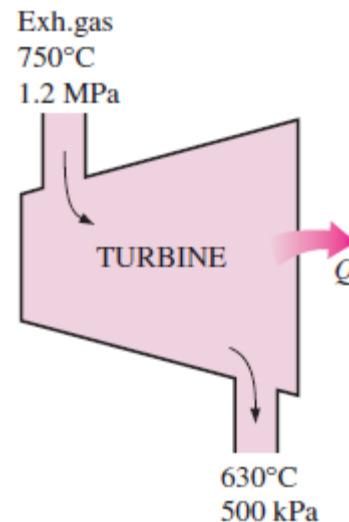
8-135 Steam is to be condensed in the condenser of a steam power plant at a temperature of 60°C with cooling water from a nearby lake that enters the tubes of the condenser at 15°C at a rate of 140 kg/s and leaves at 25°C . Assuming the condenser to be perfectly insulated, determine (a) the rate of condensation of the steam and (b) the rate of exergy destruction in the condenser. *Answers: (a) 2.48 kg, (b) 694 kW*

8-136 A well-insulated heat exchanger is to heat water ($c_p = 4.18 \text{ kJ/kg} \cdot ^\circ\text{C}$) from 25°C to 60°C at a rate of 0.4 kg/s . The heating is to be accomplished by geothermal water ($c_p = 4.31 \text{ kJ/kg} \cdot ^\circ\text{C}$) available at 140°C at a mass flow rate of 0.3 kg/s . The inner tube is thin-walled and has a diameter of 0.6 cm . Determine (a) the rate of heat transfer and (b) the rate of exergy destruction in the heat exchanger.

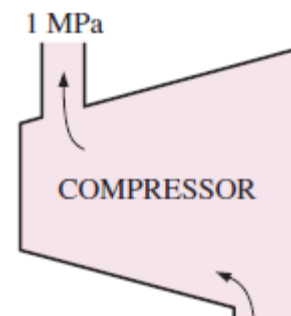


8-139 In a dairy plant, milk at 4°C is pasteurized continuously at 72°C at a rate of 12 L/s for 24 h/day and 365 days/yr . The milk is heated to the pasteurizing temperature by hot water heated in a natural gas-fired boiler having an efficiency of 82 percent . The pasteurized milk is then cooled by cold water at 18°C before it is finally refrigerated back to 4°C . To save energy and money, the plant installs a regenerator that has an effectiveness of 82 percent . If the cost of natural gas is $\$1.04/\text{therm}$ ($1\text{ therm} = 105,500\text{ kJ}$), determine how much energy and money the regenerator will save this company per year and the annual reduction in exergy destruction.

8-140 Combustion gases enter a gas turbine at 750°C and 1.2 MPa at a rate of 3.4 kg/s and leave at 630°C and 500 kPa . It is estimated that heat is lost from the turbine at a rate of 30 kW . Using air properties for the combustion gases and assuming the surroundings to be at 25°C and 100 kPa , determine (a) the actual and reversible power outputs of the turbine, (b) the exergy destroyed within the turbine, and (c) the second-law efficiency of the turbine.



8-141 Refrigerant-134a enters an adiabatic compressor at 160 kPa superheated by 3°C, and leaves at 1.0 MPa. If the compressor has a second-law efficiency of 80 percent, determine (a) the actual work input, (b) the isentropic efficiency, and (c) the exergy destruction. Take the environment temperature to be 25°C. *Answers: (a) 49.8 kJ/kg, (b) 0.78, (c) 9.95 kJ/kg*



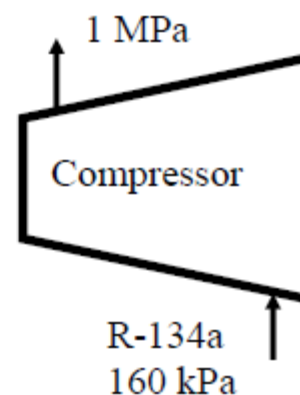
Analysis (a) The properties of the refrigerant at the inlet of the compressor are (Tables A-11 through A-13)

$$T_{\text{sat}@160 \text{ kPa}} = -15.60^\circ\text{C}$$

$$\left. \begin{array}{l} P_1 = 160 \text{ kPa} \\ T_1 = (-15.60 + 3)^\circ\text{C} \end{array} \right\} \begin{array}{l} h_1 = 243.60 \text{ kJ/kg} \\ s_1 = 0.95153 \text{ kJ/kg}\cdot\text{K} \end{array}$$

The enthalpy at the exit for if the process was isentropic is

$$\left. \begin{array}{l} P_2 = 1 \text{ MPa} \\ s_2 = s_1 = 0.95153 \text{ kJ/kg}\cdot\text{K} \end{array} \right\} h_{2s} = 282.41 \text{ kJ/kg}$$



The expressions for actual and reversible works are

$$w_a = h_2 - h_1 = (h_2 - 243.60) \text{ kJ/kg}$$

$$w_{\text{rev}} = h_2 - h_1 - T_0(s_2 - s_1) = (h_2 - 243.60) \text{ kJ/kg} - (25 + 273 \text{ K})(s_2 - 0.95153) \text{ kJ/kg}\cdot\text{K}$$

Substituting these into the expression for the second-law efficiency

$$\eta_{\text{II}} = \frac{w_{\text{rev}}}{w_a} \longrightarrow 0.80 = \frac{h_2 - 243.60 - (298)(s_2 - 0.95153)}{h_2 - 243.60}$$

The exit pressure is given (1 MPa). We need one more property to fix the exit state. By a trial-error approach or using EES, we obtain the exit temperature to be 60°C. The corresponding enthalpy and entropy values satisfying this equation are



$$h_2 = 293.36 \text{ kJ/kg}$$

$$s_2 = 0.98492 \text{ kJ/kg}\cdot\text{K}$$

Then,

$$w_a = h_2 - h_1 = 293.36 - 243.60 = \mathbf{49.76 \text{ kJ/kg}}$$

$$w_{\text{rev}} = h_2 - h_1 - T_0(s_2 - s_1) = (293.36 - 243.60) \text{ kJ/kg} - (25 + 273 \text{ K})(0.98492 - 0.9515) \text{ kJ/kg} \cdot \text{K} = 39.81 \text{ kJ/kg}$$

(b) The isentropic efficiency is determined from its definition

$$\eta_s = \frac{h_{2s} - h_1}{h_2 - h_1} = \frac{(282.41 - 243.60) \text{ kJ/kg}}{(293.36 - 243.60) \text{ kJ/kg}} = \mathbf{0.780}$$

(b) The exergy destroyed in the compressor is

$$x_{\text{dest}} = w_a - w_{\text{rev}} = 49.76 - 39.81 = \mathbf{9.95 \text{ kJ/kg}}$$

8-142 Water enters a pump at 100 kPa and 30°C at a rate of 1.35 kg/s, and leaves at 4 MPa. If the pump has an isentropic efficiency of 70 percent, determine (a) the actual power input, (b) the rate of frictional heating, (c) the exergy destruction, and (d) the second-law efficiency for an environment temperature of 20°C.

8-143 Argon gas expands from 3.5 MPa and 100°C to 500 kPa in an adiabatic expansion valve. For environment conditions of 100 kPa and 25°C, determine (a) the exergy of argon at the inlet, (b) the exergy destruction during the process, and (c) the second-law efficiency.

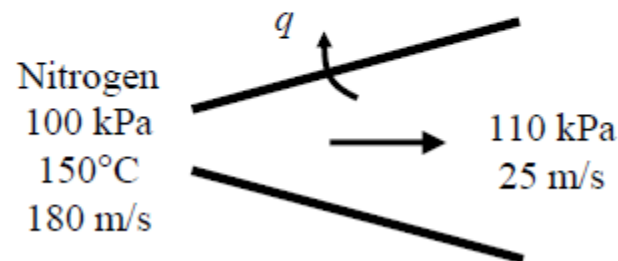


8-144 Nitrogen gas enters a diffuser at 100 kPa and 150°C with a velocity of 180 m/s, and leaves at 110 kPa and 25 m/s. It is estimated that 4.5 kJ/kg of heat is lost from the diffuser to the surroundings at 100 kPa and 27°C. The exit area of the diffuser is 0.06 m². Accounting for the variation of the specific heats with temperature, determine (a) the exit temperature, (b) the rate of exergy destruction, and (c) the second-law efficiency of the diffuser. *Answers: (a) 161°C, (b) 5.11 kW, (c) 0.892*

Analysis (a) For this problem, we use the properties from EES software. Remember that for an ideal gas, enthalpy is a function of temperature only whereas entropy is functions of both temperature and pressure. At the inlet of the diffuser and at the dead state, we have

$$\left. \begin{array}{l} T_1 = 15^\circ\text{C} = 423 \text{ K} \\ P_1 = 100 \text{ kPa} \end{array} \right\} \begin{array}{l} h_1 = 130.08 \text{ kJ/kg} \\ s_1 = 7.2006 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} T_1 = 300 \text{ K} \\ P_1 = 100 \text{ kPa} \end{array} \right\} \begin{array}{l} h_0 = 1.93 \text{ kJ/kg} \\ s_0 = 6.8426 \text{ kJ/kg} \cdot \text{K} \end{array}$$



An energy balance on the diffuser gives

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} + q_{\text{out}}$$

$$130.08 \text{ kJ/kg} + \frac{(180 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) = h_2 + \frac{(25 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) + 4.5 \text{ kJ/kg}$$

$$\longrightarrow h_2 = 141.47 \text{ kJ/kg}$$

The corresponding properties at the exit of the diffuser are

$$\left. \begin{array}{l} h_2 = 141.47 \text{ kJ/kg} \\ P_2 = 110 \text{ kPa} \end{array} \right\} \begin{array}{l} T_2 = \mathbf{160.9^\circ\text{C}} = 433.9 \text{ K} \\ s_2 = 7.1989 \text{ kJ/kg} \cdot \text{K} \end{array}$$

(b) The mass flow rate of the nitrogen is determined to be

$$\dot{m} = \rho_2 A_2 V_2 = \frac{P_2}{RT_2} A_2 V_2 = \frac{110 \text{ kPa}}{(0.2968 \text{ kJ/kg.K})(433.9 \text{ K})} (0.06 \text{ m}^2)(25 \text{ m/s}) = 1.281 \text{ kg/s}$$

The exergy destruction in the nozzle is the exergy difference between the inlet and exit of the diffuser

$$\begin{aligned} \dot{X}_{\text{dest}} &= \dot{m} \left[h_1 - h_2 + \frac{V_1^2 - V_2^2}{2} - T_0 (s_1 - s_2) \right] \\ &= (1.281 \text{ kg/s}) \left[(130.08 - 141.47) \text{ kJ/kg} + \frac{(180 \text{ m/s})^2 - (25 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) \right. \\ &\quad \left. - (300 \text{ K})(7.2006 - 7.1989) \text{ kJ/kg.K} \right] = \mathbf{5.11 \text{ kW}} \end{aligned}$$

(c) The second-law efficiency for this device may be defined as the exergy output divided by the exergy input:

$$\begin{aligned} \dot{X}_1 &= \dot{m} \left[h_1 - h_0 + \frac{V_1^2}{2} - T_0 (s_1 - s_0) \right] \\ &= (1.281 \text{ kg/s}) \left[(130.08 - 1.93) \text{ kJ/kg} + \frac{(180 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) - (300 \text{ K})(7.2006 - 6.8426) \text{ kJ/kg.K} \right] \\ &= 47.35 \text{ kW} \\ \eta_{\text{II}} &= \frac{\dot{X}_2}{\dot{X}_1} = 1 - \frac{\dot{X}_{\text{dest}}}{\dot{X}_1} = 1 - \frac{5.11 \text{ kW}}{47.35 \text{ kW}} = \mathbf{0.892} \end{aligned}$$

Chapter 8

End of the Problems